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HoRNS-CNN model: an energy-efficient fully homomorphic residue number system convolutional neural network model for privacy-preserving classification of dyslexia neural-biomarkers

Opeyemi Lateef Usman^{1,2*} , Ravie Chandren Muniyandi¹ , Khairuddin Omar³ , Mazlyfarina Mohamad⁴ , Ayoade Akeem Owoade² and Morufat Adebola Kareem²

Abstract

Recent advancements in cloud-based machine learning (ML) now allow for the rapid and remote identification of neural-biomarkers associated with common neuro-developmental disorders from neuroimaging datasets. Due to the sensitive nature of these datasets, secure deep learning (DL) algorithms are essential. Although, fully homomorphic encryption (FHE)-based methods have been proposed to maintain data confidentiality and privacy, however, existing FHE deep convolutional neural network (CNN) models still face some issues such as low accuracy, high encryption/decryption latency, energy inefficiency, long feature extraction times, and significant cipher-image expansion. To address these issues, this study introduces the HoRNS-CNN model, which integrates the energy-efficient features of the residue number system FHE scheme (RNS-FHE scheme) with the high accuracy of pre-trained deep CNN models in the cloud for efficient, privacy-preserving predictions and provide some proofs of its energy efficiency and homomorphism. The RNS-FHE scheme's FPGA implementation includes embedded RNS pixel-bitstream homomorphic encoder/decoder circuits for encrypting 8-bit grayscale pixels, with cloud CNN models performing remote classification on the encrypted images. In the HoRNS-CNN architecture, the ReLU activation functions of deep CNNs were initially trained for stability and later adapted for homomorphic computations using a Taylor polynomial approximation of degree 3 and batch normalization to achieve high accuracy. The findings show that the HoRNS-CNN model effectively manages cipher-image expansion with an asymptotic complexity of $O(n^3)$, offering better performance and faster feature extraction compared to its peers. The model can predict 400,000 neural-biomarker features in one hour, providing an effective tool for analyzing neuroimages while ensuring privacy and security.

Keywords Privacy-preserving methods, Fully homomorphic encryption, Residue number system, Neuroimages, Neural-biomarkers, Dyslexia prediction

*Correspondence:

Opeyemi Lateef Usman
usmanol@tasued.edu.ng

Full list of author information is available at the end of the article

1 Introduction

Maintaining privacy in the classification of sensitive datasets, particularly in the field of medicine, is becoming increasingly crucial with the deployment of powerful machine learning (ML) algorithms in the cloud. Cloud-based ML platforms such as Google Cloud Machine Learning Engine (Google Prediction API), AWS Machine Learning, Microsoft Azure Machine Learning Studio, Ersatz Labs, and GraphLab are examples of platforms that support DL algorithms' training and simulation using the "*model as a service*" paradigm [1–4]. With the increasing generation of neuroimages such as MRI scans, cloud-based ML platforms are becoming more innovative and relevant. While users can develop new algorithms or use pre-trained algorithms, the cloud server managing these platforms cannot be fully trusted. Therefore, implementing secure DL models on these platforms is crucial for detecting and addressing potential security threats and vulnerabilities, as well as preventing the theft of sensitive information [5]. Security vulnerabilities are defects, weaknesses, or errors identified in a security system that can be exploited by attackers to compromise the security of a network [6–8]. Therefore, ensuring the privacy of patients' medical records is crucial, particularly when these records contain sensitive neural-biomarker features. Although remote classification can enhance the quality and speed of medical decision-making while reducing costs, the privacy of these datasets and their critical features may be at risk of exposure to different forms of threats and attacks through DL hyperparameters [9, 10].

Current methods for ensuring privacy and secure transmission of medical images rely on existing crypto-based and cryptographic algorithms that utilize pixel confusion and diffusion processes [11, 12]. However, these processes can damage the images' key regions of interest (ROI) and make the corresponding cipher-images unsuitable for remote classification. Moreover, these algorithms lack the capability to efficiently process large and complex datasets (big data), leading to increased encryption latency, heightened circuit complexity, asymptotic inefficiency, and higher energy consumption. Additionally, current state-of-the-art homomorphic CNN models exhibit a trade-off between energy efficiency and classification performance. This trade-off arises from several factors, including large ciphertext sizes and intricate polynomial arithmetic, computationally expensive modular arithmetic transformations, increased CNN depth, a higher number of filters or larger kernel sizes, and the execution of homomorphic inference on cloud-based GPU/TPU environments when processing extensive neuroimaging datasets. These factors collectively contribute to elevated energy consumption while simultaneously

compromising classification efficiency. Recent studies such as [13–15] have suggested the use of mathematically-efficient methods that combine the homomorphic capabilities of the FHE scheme with the high accuracy capability of CNN models to address the above limitations. Nevertheless, to implement these methods with reduced latency and low circuit complexity, an energy-efficient privacy-preserving approach is required [1] to overcome the shortcomings of current algorithms while still maintaining data privacy.

Homomorphic Encryption (HE) delegates encrypted data processing to an untrusted cloud server without revealing the secret key, and the decrypted output matched the raw data. It is regarded as one of the effective ways to protect cloud-based user data [16]. Craig Gentry [17] proposed the first FHE scheme to solve cryptographic problems based on ideal lattices. Without a decryption key, arbitrary functions can be computed over encrypted data. FHE expands the range of computations that can be employed to process encrypted data homomorphically. Dowlin et al. [13] developed a high-accuracy CryptoNet by using a FHE scheme to construct a shallow-depth homomorphic CNN model. Chabanne et al. [18] and Hesamifard et al. [19] expanded CryptoNet by integrating polynomial approximation activation functions that utilize batch normalization for stability. The above studies face significant challenges in designing and implementing privacy-preserving methods using deep CNN models. The majority of FHE-CNN models proposed have only been evaluated on high-resolution images (512×512 pixels), while others implemented with mid-range resolution images perform poorly in terms of both computational time and memory utilization [15]. Thus, evaluating FHE-CNN for practical medical image privacy-preserving classification necessitates a strategy for reducing computational time, memory usage, and design complexity [20]. On the contrary, existing FHE-CNN methods for medical images are characterized by high computational cost, latency, asymptotic inefficiency, and large cipher-image expansion problems. Due to the large overhead, extracting the most relevant features from cipher-images takes a long time. While inference on plain networks takes a few minutes to complete, it takes hours on encrypted networks [19, 21]. Furthermore, the depth limitation of homomorphic operations prevents the use of some CNN activation functions [18] such as ReLU function. This necessitates their replacement with low-complexity approximation functions. Therefore, the practical implementation of a CNN-based privacy-preserving classification method for neuroimages will require the development of a mechanism for improving the biological interpretation of inherent neural-biomarker features, the design of an energy-efficient FHE

scheme for the image privacy, hyperparameter optimization in the cipher domain [10], and adaptation of CNN model to unlimited depth homomorphic computations during network training, thus justifying the need for this study.

This study expands on the authors' previously published conference paper (See Reference [10]) by elaborating the design paradigm of the HoRNS-CNN model, proof of energy efficiency, demonstrating the homomorphic property, and the asymptotic efficiency of the RNS-FHE scheme. For a detailed security analysis and evaluation of the proposed scheme, refer to the study by Usman et al. [8]. HoRNS-CNN concatenates the energy-efficient RNS-FHE scheme and pre-trained deep CNN networks. The HoRNS-CNN model extends "CryptoNets" and "CryptoDL" with the energy-efficient capability of a secure RNS-FHE scheme. The study's objective is, therefore, to design an energy-efficient secured RNS-FHE scheme for a privacy-preserving classification with deep CNN models for dyslexia neural-biomarkers, that outperforms current state-of-the-art methods in prediction accuracy and computation time. The study also intends to provide some proofs of energy efficiency, homomorphism, and asymptotic efficiency of the proposed HoRNS-CNN model.

1.1 Motivation and contributions

This work is motivated by the desire to improve the security, efficiency, and accuracy of current state-of-the-art deep CNN classifiers for neural-biomarkers classification using a case study of dyslexia neural-biomarker. This was achieved by designing an energy-efficient RNS secure deep CNN algorithm and a model called HoRNS-CNN. The proposed model integrates the energy-efficient feature of RNS-FHE with the high accuracy of cloud-based pre-trained deep CNN models to enable efficient privacy-preserving prediction of neural-biomarker features inherent in neuroimages. The main contribution of this work, therefore, is the development of an energy-efficient secure RNS-FHE privacy-preserving classification framework that integrates with cloud-based deep CNNs for neural-biomarkers prediction, thus, forming the design principle for HoRNS-CNN model. Other contributions of this study include formalization and simulation of RNS-FHE scheme and its security evaluation, demonstration of its homomorphic property, and its asymptotic efficiency, and lastly, its cloud-based implementation. The specific contributions of this study are, therefore, summarized thus:

- i. One of the key contributions of this research is the development of a privacy-preserving classification conceptual framework, called RNS-FHE, which is

both energy-efficient and secure. This framework is seamlessly integrated with cloud-based deep CNNs to enable prediction of neural-biomarkers. Consequently, this framework forms the fundamental design principle for the HoRNS-CNN model.

- ii. This study involves the mathematical formalization and FPGA simulation of the energy-efficient RNS-FHE scheme, which subsequently leads to the development of homomorphic encoder and decoder circuits for neuroimage data encryption and decryption. These circuits are otherwise, referred to as RNS pixel-bitstream homomorphic circuits. These circuits are specifically designed to encrypt and decrypt each 8-bit grayscale pixel of the normalized neuroimages while retaining the crucial features necessary for discriminating dyslexia neural-biomarker features.
- iii. The energy efficiency of the proposed RNS-FHE scheme is evaluated through an analysis of the critical path latency of the RNS pixel-bitstream homomorphic encoder and decoder circuits that were designed. The results indicate a significant reduction in homomorphic computational latency of these circuits, particularly for values of $n \geq 3$, when compared to the current state-of-the-art equivalent circuits. Additionally, the FPGA implementation of the RNS pixel-bitstream homomorphic circuits demonstrated an energy savings of up to 42.4% compared to ROM-based implementations. Furthermore, the decryption of cipher-neuroimages was achieved with a polynomial asymptotic time-complexity of $O(n^3)$.
- iv. This study demonstrates the robustness and security of the RNS-FHE scheme against various statistical attacks, including histogram analysis, brute-force attacks, and correlation coefficient analysis. The proposed scheme is shown to be highly sensitive to even slight changes in the key parameters, further reinforcing its security strength [8].
- v. Finally, the HoRNS-CNN architecture utilized pre-trained cloud CNN models, which resulted in efficient performance for classification and feature extraction tasks. Importantly, the models achieved this performance while maintaining the confidentiality of the neuroimaging dataset, ensuring that their contents remained undisclosed. The classification accuracy reached an impressive 91.4%, indicating the effectiveness of the approach.

The structure of this paper is as follows: Sect. 2 reviews recent studies on privacy-preserving classification, HE, and the RNS, which serve as the theoretical framework for this study. Section 3 presents materials and methods.

Sections 4 and 5 describe the experiment setup, simulation, results, findings, and discussion, while Sect. 6 concludes the work with some practical implications.

2 Literature review

2.1 Privacy-preserving classification and homomorphic encryption

Privacy-preserving classification is a concept that describes a scenario in which cloud-based ML models perform direct arbitrary computations and make inferences on highly confidential data in the encrypted (cipher) form. Studies have suggested privacy-preserving classification methods based on either perceptual encryption algorithms [1, 22–25] or homomorphic encryption (HE) algorithms [2, 15, 19, 26, 27].

Perceptual encryption algorithms-based privacy-preserving methods conceal visual information by making a security trade-off to meet other requirements such as low processing demand, bitstream compliance, and signal processing in the cipher form. This has been suggested for various block-based encryption-then-compression schemes [28] with artificial neural networks (ANNs), support vector machines (SVM), decision trees (DT), and Naïve Bayes applications [1, 24, 29]. These methods have the advantage of having a low computational cost and the ability to provide encrypted data that is resistant to various types of noise and errors. Furthermore, to be adaptable to cloud storage and network sharing, some of them intend to consider both security and efficient compression [22]. However, cipher-image classification accuracy with these methods is significantly lower than with plain images, and they do not allow for data augmentation [30].

In HE-based privacy-preserving methods, certain arbitrary computations such as additions, multiplications, and XOR-functions can be performed directly on the cipher data; and this is currently being used in image classification tasks involving CNN and public datasets ranging from CIFAR-10 [19] to MNIST [13, 14] and, more recently, to medical images [15, 31]. Despite the benefits of HE schemes, they do have some important general drawbacks. The first among them is the limitation to only integer message space [32]. Integers are used in almost all HE schemes. Thus, before encrypting data, it must be converted to integers. The size of the ciphertext is the second constraint. Encryption dramatically expands the size of the plain message [8, 16]. Another significant constraint is related to noise. With each homomorphic computation, the amount of noise in ciphertext increases. Multiplication produces a lot more noise than addition and subtraction [33–35]. The amount of noise should always

be kept below a pre-defined threshold level. The final limitation is the absence of a division operation. It is worth noting that complex functions in CNN model, such as ReLU activations and max-pooling functions, only allow for a limited depth of addition and multiplication operations on encrypted data. Because of this limitation, CNN models are incompatible with HE in their original forms [36]. In addition, homomorphic CNN approaches are difficult to scale to image resolution, resource efficiency, and latency requirements [15]. For the majority of studies that employ homomorphic CNN for the medical image classification task, images with resolutions greater than or equal to the size 512×512 pixels are considered. A large amount of time and memory space is required to process images of low to medium resolution range [14, 15]. Consequently, practical evaluation of privacy-preserving homomorphic CNN for the classification of low-to-medium resolution medical images requires strategies that significantly reduce computational complexity, improve prediction accuracy, decrease memory usage, and feature extraction time [10, 14], thus justifying the need for this study.

Homomorphic encryption (HE) is not a new concept in the field of cryptography. Its properties have been known since 1978, following the development of the Rivest, Shamir, and Adleman [37] asymmetric algorithm known as RSA. Craig Gentry [17], an IBM researcher, presented his doctoral thesis on a cryptosystem that combines aspects of prior homomorphic algorithms in 2009, spawning the concept of a FHE scheme. Gentry's original scheme was inefficient, but subsequent studies produced significantly more practical schemes, leading to the classification of FHE schemes into three developmental eras [38–40]: first-era FHE is based on ideal lattices; the second era is based on learning with error (LWE) and ring learning with error (RLWE) problems; and the third era is Gentry, Sahai, and Waters (GSW) scheme, which is also based on RLWE. However, each of these FHE scheme eras needs to be improved further to be useful in real-world applications [16] including privacy-preserving classifications. FHE scheme refers to the ability of a scheme to perform an unlimited depth of arbitrary homomorphic computations on cipher data. Equation (1–8) provide some definitions of important concepts in HE. Formally, the HE scheme is defined as quadruple.

$$\varepsilon = (\text{KeyGen}, \text{Enc}, \text{Dec}, \text{Eval}) \quad (1)$$

where:

i. *KeyGen* is a key generation algorithm that takes security parameter λ as input and outputs a finite set of keys

(pk, sk, ek) where pk =public key, sk =secret key, and ek =evaluation key.

ii. Enc is the encryption algorithm that converts plaintext to ciphertext as follows:

$$c \leftarrow Enc(m, pk) \text{ where } m \text{ is the plaintext and } c \text{ is the ciphertext} \quad (2)$$

iii. Dec is the decryption algorithm that converts ciphertext to plaintext as follows:

$$m \leftarrow Dec(c, sk) \text{ where } c \text{ is the ciphertext and } m \text{ is the plaintext} \quad (3)$$

iv. $Eval$ is the homomorphic evaluation algorithm that evaluates the result of a computation f on ciphertexts $c_1, c_2, \dots, c_n$ using evaluation key ek as follows:

$$c_f \leftarrow Eval(f, c_1, c_2, \dots, c_n, ek), \text{ for all } c_i \in C \quad (4)$$

f is a function that maps elements of a ciphertext set $C = (c_1, c_2, \dots, c_n)$ onto the elements of a message set $M = (m_1, m_2, \dots, m_n)$. If f is represented as an *arithmetic circuit* or *Boolean circuit*, then the scheme is said to be *circuit-based* homomorphic, otherwise, if f is defined as a mathematical function, then the scheme is *non-circuit based* homomorphic [40].

v. An encryption scheme ϵ is homomorphic if $\forall c_i \in C$, the evaluation algorithm outputs.

$$c_f \leftarrow Eval(f, c_1, c_2, \dots, c_n, ek) \text{ and } f(m_1, m_2, \dots, m_n) = Dec(c_f, sk) \quad (5)$$

vi. **Additive property:** A homomorphic encryption scheme ϵ is additive homomorphic if

$$Enc(c_1 \oplus c_2) \leftarrow Enc(c_1) \oplus Enc(c_2) \text{ and}$$

$$Enc\left(\sum_{i=1}^n c_i\right) = \prod_{i=1}^n Enc(c_i) \text{ where operator } \oplus \text{ is addition} \quad (6)$$

vii. **Multiplicative property:** A homomorphic encryption scheme ϵ is multiplicative homomorphic if

$$Enc(c_1 \oplus c_2) \leftarrow Enc(c_1) \oplus Enc(c_2) \text{ and}$$

$$Enc\left(\prod_{i=1}^n c_i\right) = \prod_{i=1}^n Enc(c_i) \text{ where operator } \oplus \text{ is multiplication} \quad (7)$$

viii. **C-Homomorphism:** Let C be a class of functions. A scheme ϵ is C -homomorphic if for every function in C , the $Eval$ algorithm of ϵ outputs such that Eq. (8) holds:

$$c \leftarrow Eval(f, c_1, c_2, \dots, c_i, ek) \text{ and } f(\pi_1, \pi_2, \dots, \pi_i) = Dec(c_f, key_2) \quad (8)$$

2.2 Residue number system (RNS) and homomorphic encryption

The residue number system (RNS) is a modular, non-weighted, carry-free number system with parallelism features. These features are used in digital signal processing (DSP) applications such as filtering, Fourier transformations (discrete and fast), and cryptography [41]. Equations (9-13) provide formal definitions of RNS in relation to homomorphic computations. Let S denote the moduli set for a given RNS scheme.

$$S = \{m_1, m_2, \dots, m_n\}, GCD(m_i, m_j) = 1 \text{ for } i \neq j, \quad (9)$$

The dynamic range (DR), M of the RNS scheme is computed as:

$$M = \prod_{i=1}^n m_i, \quad (10)$$

Any integer X in the range $[0, M - 1]$ element of the residue class $\mathbb{Z}_M$ has a unique representation in the RNS using n -tuple of x_i as follows:

$$X \xrightarrow{RNS} (x_1, x_2, \dots, x_n), \text{ where } x_i = X \bmod m_i = |X|_{m_i}, \quad (11)$$

Assuming variables X and Y are integer values bounded within the ring $\mathbb{Z}_M$, then.

$$X \oplus Y \xrightarrow{RNS} (x_1 \oplus y_1, x_2 \oplus y_2, \dots, x_n \oplus y_n), \quad (12)$$

where $\oplus$ is a set of binary operators $\{+, -, \times\}$,

This implies that the above RNS representation is homomorphic w.r.t. *addition*, *subtraction*, and *multiplication*, all of which are integral operations in RNS [33]. In other words, the above expression can be written as:

$$\begin{aligned} X \pm Y &\xrightarrow{RNS} (x_1 \pm y_1, x_2 \pm y_2, \dots, x_n \pm y_n) \\ &= |x_i \pm y_i|_{m_i}, \\ X \times Y &\xrightarrow{RNS} (x_1 \times y_1, x_2 \times y_2, \dots, x_n \times y_n) \\ &= |x_i \times y_i|_{m_i}. \end{aligned} \quad (13)$$

where x_i and y_i are the cipher forms of integer X and Y in RNS homomorphic cryptosystem.

The moduli set selection is a critical factor in the successful design of an RNS-based cryptosystem. In the field of cryptography, early studies proposed either arbitrary moduli set [42] or special moduli set of the form $\{2n \pm k, 2^n \pm k, 3n \pm k, 3^n \pm k\}$ for designing RNS cryptographic systems [16, 43]. For instance, Weyori et al. [44] proposed a digital image encryption scheme using an arbitrary moduli set $\{2n + 2, 2n + 1, 2n\}$ whereas Alhasan & Gholagade [43] and Usman et al. [45] developed a

binary-to-residue (B/R) encoder and decoder circuits for digital image encryption using the popular traditional power-of-two moduli set $\{2^n - 1, 2^n, 2^n + 1\}$. Although, the traditional moduli set is well-known for simplifying the computations required to implement reverse converters (decoders), the modulo- $(2^n + 1)$ channel remains the system's bottleneck. Its arithmetic circuit has the longest delay among the three channels. Therefore, the number of modulo of the form $(2^k + 1)$ need to be minimized [46] to improve efficiency.

In this study, an attempt was made to hybridize the energy-efficiency property of the RNS-FHE scheme with the high prediction accuracy of pre-trained deep CNN models to develop the HoRNS-CNN model for securely predicting dyslexia neural-biomarker features. The HoRNS-CNN model has a low circuit complexity and the ability to manage cipher expansion, resulting in a fast feature extraction time. To reduce the computational overhead of modulo of the form $(2^k + 1)$, this study replaces modulo- $(2^n + 1)$ of previous studies with modulo- $(2^{n+1} - 1)$. This allows us to simplify homomorphic computations while also improving the energy-efficiency of the proposed RNS-FHE scheme.

3 Materials and methods

3.1 Data collection and pre-processing

The MRI datasets used in this study were obtained from publicly available databases such as OpenNeuro and Beijing Normal University's Connectivity-based Brain Imaging Research Database (C-BIRD). The datasets were generated by experimenting with different scanners' field strengths and acquisition parameters. Usman et al. [47] present detailed demographic information about the subjects, as well as scan acquisition methods, parameters, and protocols for the two datasets that can be accessed via:

- i. <https://openneuro.org/datasets/ds002236/versions/1.1.0> and.
- ii. http://dx.doi.org/https://doi.org/10.15387/fcp_indi.corr.bnu1

The OpenNeuro dataset contains MRI scans from 91 subjects (aged 8.7–15.5 years) and investigates the neural mechanisms and development of lexical processing through task-based fMRI of rhyming, spelling, and semantic judgment tasks in both the auditory and visual modalities [48]. The dataset was stored under the Brain Imaging Data Structure (BIDS) recent version 1.3.0 specifications [49] and contains test scores from six different standardized psycho-educational tests used to assess an individual's reading and cognitive abilities. The psycho-educational tests include the Woodcock-Johnson

III (WJ-III), Wide Range Achievement Test: Revision 3 (WRAT-3), Wechsler Abbreviated Scale of Intelligence (WASI), Comprehensive Test of Phonological Processing (CTOPP), Test of Word Reading Efficiency (TOWRE), and Peabody Picture Vocabulary Test: 3rd Edition (PPVT-III). The CTOPP composite scores were used to categorize the subjects because they assess the phonological processing elements for dyslexia classification, are appropriate for individuals aged 4 to 24, and are simple to interpret [50]. The test measures three types of reading-related phonological processing elements: phonological awareness, phonological memory, and rapid naming. Dyslexia can be caused by a deficit in the composite score of any of the above elements. However, we used the '*double-deficit dyslexia*' approach to divide the subjects into two groups: dyslexic and non-dyslexic. In this study, '*double-deficit dyslexia*' refers to composite scores that are lower than the average of 90 (i.e., composite score < 90) in both phonological awareness and either phonological memory or rapid naming, or all three elements. From the above dataset, 18 subjects have deficits in both phonological awareness and rapid naming, 16 subjects have deficits in both phonological awareness and phonological memory. Therefore, 34 subjects have deficiencies in at least two of the three elements. The Supplementary1.csv file contains more information about this categorization and its interpretation.

The C-BIRD dataset contains MRI scans from 57 healthy young volunteer subjects (aged 19–30 years) who completed two resting-state fMRI scan sessions over 6 weeks (40.94 ± 4.51 days). Every subject in the study was right-handed and had no history of neurological or psychiatric disorders. The dataset is also under the recent version of BIDS specifications. The State Key Laboratory of Cognitive Neuroscience and Learning at Beijing Normal University Institutional Review Board has approved the data acquisition protocols. To avoid duplication of data for the same subject, only the first session scans were used.

Table 1 summarizes the different types of MRI scans found in the two datasets discussed above. We used the first five modules of the 'Dyslexia neural-biomarkers interpretability and classification mechanism' (See Fig. 1

Table 1 Types of neuroimaging datasets in the used databases

Database	Dyslexic class	Normal class	Total	MRI type
OpenNeuro	34	57	91	T1-w SPGR sMRI, T2-w BOLD task-based fMRI
C-BIRD	-	57	57	T1-w and T2-w resting-state fMRI, DTI
Total	34	114	148	

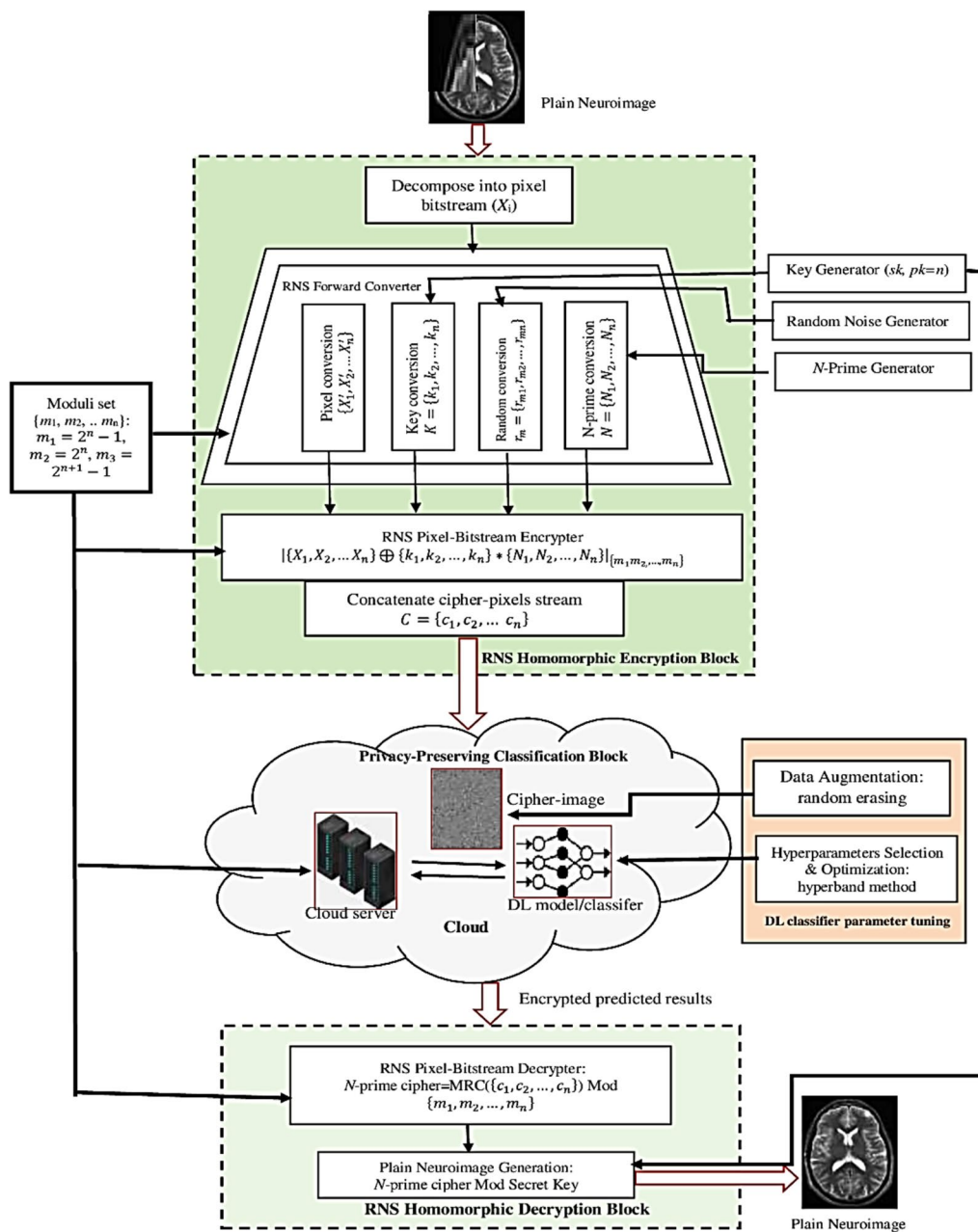


Fig. 1 Proposed energy-efficient privacy-preserving classification framework (Updated version of Usman et al. [10])

of Usman et al. [47]) to pre-process the dataset to achieve homogeneous image intensity and smoothness (Also summarized in the Algorithm 3.1 and 3.2). The dataset was then segmented into three major brain tissues using the Markov random field segmentation algorithm [51]. This algorithm was implemented in the computational anatomy toolbox version 12 (CAT12), which is a MATLAB-compatible extension of the statistical parametric mapping (SPM12) package. The FSL FNIRT software tool was then used to perform a non-rigid image registration

into the Montreal Neurological Institute (MNI152) brain template [52]. The ROIs for this study is the cognitive/phonological features extracted from grey matter (GM) tissue from three brain regions. They are the left superior temporal gyrus (LSTG), the left occipital temporal gyrus (LOTG), and the lateral cerebellum (LC). LSTG is linked to language (speech), visual, auditory, and motor-skill development; LOTG is linked to visual development, and LC is linked to motor-skill development. Therefore, any abnormalities in the structure or activation of these

regions result in a wide range of neuro-developmental disabilities, including dyslexia [53]. The Supplementary2.docx file contains samples from the two databases.

Algorithm 3.1. Proposed Gaussian smoothing algorithm with isotropic kernel for neuroimages smoothness

smoothness

Input: The neuroimage set I_i ($i=1, 2, \dots, n$) such that $I_i \subset I$, kernel size sz , Standard deviation σ , $\pi=3.1412$

Output: Gaussian smooth neuroimages I

Procedure:

1: **Read** a neuroimage I_i //image I_i has dimension $m \times n$

Initialization:

2: $\sigma = \{0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00, 2.25, 3.0, \dots, 5.00\}$

3: $sz = 4.0$ // Kernel size implies the Window size

4: compute $I = \text{double}(I_i)$

5: **For** $i=1$ to n do

6: $[x,y]=\text{meshgrid}(-sz:sz,-sz:sz)$

7: $m = \text{size}(x,1)-1$ //resize image length x

6: $n = \text{size}(y,1)-1$ //resize image height y

7: **Compute** $\text{Gaussian_exp} = -(x^2+y^2)/(2*\sigma*\sigma)$

8: **Compute** $2D_Gaussian_kernel = \exp(\text{Gaussian_exp})/(2*\pi*\sigma*\sigma)$

9: **End for**

10: **End**

Algorithm 3.2 Proposed modified histogram normalization algorithm for neuroimage intensity normalization

normalization

Input: A neuroimage set: $I_i \in I$ where ($i = 1, 2, \dots, n$),

Coordinate: x, y ,

Low-intensity ROI and high-intensity ROI: S_1, S_2 ,

Minimum and maximum intensity level: q_{min}, q_{max}

Output: Scaled-up boundaries: lower boundary m_1^1 , upper boundary m_2^1 ,

Normalized image I_i^S, I^S

Begin

1: **For** $i=1$ to n do //This part covers the image intensity scaling.

2: Compute histogram h_i of image I_i using $g(x, y) = \frac{f(x,y)-ROI_{low}}{ROI_{high}-ROI_{low}}$;

3: Determine minimum intensity value q_{min} that corresponds to S_1 and maximum intensity values q_{max} that corresponds to S_2 where ($S_1 = ROI_{low}, S_2 = ROI_{high}$) and landmark locations m_1

and m_2 on I_i , using $h(x, y) = \frac{ROI_{high}-ROI_{low}}{q_{max}-q_{min}}(g(x, y) - q_{min}) + ROI_{low}$;

4: Perform **linear mapping** $[I_{min}^{h_i}, I_{max}^{h_i}]$ of I_i onto $[I_{min}^S, I_{max}^S]$;

5: Find the new mapped landmark m_1^1 and m_2^1 rounded mean value μ_i and μ_s ;

6: **return**($m_1^1, m_2^1, \mu_i, \mu_s$);

7: end for;

8: Compute the histogram $h_i \leftarrow (I_i, f_{ic}^i)$ of I_i using $g(x, y) = \frac{f(x,y)-ROI_{low}}{ROI_{high}-ROI_{low}}$ and $h(x, y) =$

$\frac{ROI_{high}-ROI_{low}}{q_{max}-q_{min}}(g(x, y) - q_{min}) + ROI_{low}$;

9: Perform linear mapping of a section of the scale of I_i onto the standard scale I_s Is of the standard histogram $h^S \leftarrow (I^S, f_{ic}^S)$;

10: **Map** the intensity value of every pixel $P_i \in I_i$ in order to obtain normalized image I_i^S using

normalization function of $N(x, y) = \left[\begin{array}{l} \mu_s + (g(x, y) - \mu_i) \frac{S_1 - \mu_s}{P_1 - \mu_i} \\ \mu + (g(x, y) - \mu) \frac{S_2 - \mu_s}{P_2 - \mu_i} \end{array} \right], m_1' \leq g(x, y) \leq \mu_i \leq m_2'$;

11: **End**

3.2 Proposed energy-efficient privacy-preserving classification framework

The proposed conceptual framework for the privacy-preserving classification of dyslexia neural-biomarkers prediction is based on the energy-efficient RNS-FHE scheme, as shown in Fig. 1. This framework consists of three major blocks: RNS homomorphic encryption, privacy-preserving classification, and RNS homomorphic decryption as illustrated in [10]. In the initial phase, the pre-processed neuroimages are decomposed into bitstreams representing their pixel values, each ranging from [0, 255]. These pixel bitstreams are subsequently transmitted to the RNS homomorphic encryption block for further processing within the FHE scheme, as outlined below:

RNS homomorphic encryption block: In this block, neuroimages I_i , are split into pixel-bitstreams X_i ($X_i \in I_i$), each of which has an 8-bit grayscale binary value in the range [0–255]. For the entire scheme, a pair of keys is generated: a very large N -prime integer N_i , as the secret key (sk) and the value of n ($n \geq 3$) as the public key (pk) for ascertaining the moduli set's integer values. As the noise, the random integer value r_{m_i} , in a ring $\mathbb{Z}_n$ is also generated. In addition, the value of n was used to produce a set of pairwise relatively prime integers that represent the moduli set. Three-channel power-of-two special moduli set $m_i = \{2^n - 1, 2^n, 2^{n+1} - 1\}$ was used to integrate an energy-efficiency property into the scheme. The computational complexity of modulo- $(2^n + 1)$ in the most widely used moduli set $\{2^n - 1, 2^n, 2^n + 1\}$ has been decreased in this regard by replacing it with a low computational complexity modulo- $(2^{n+1} - 1)$. This allows the RNS-FHE scheme to perform medium dynamic range (DR) homomorphic computation of $(3n + 1)$ -bit. The RNS forward converter computes the residue x_i , of each pixel-bitstream from X_i , n , r_{m_i} , and N_i w.r.t. m_i and transmits the output to the RNS pixel-bitstream homomorphic encrypter block which generates corresponding cipher-pixels c_i . The c_i elements are finally concatenated to generate the cipher-images C_i where $i = 1, 2, 3, \dots$. This block's outputs were then forwarded to cloud DL algorithms for privacy-preserving classification.

Privacy-preserving classification block: The cloud stores the pre-trained DL models that make predictions on encrypted neuroimages (cipher-images), C_i . While the previous block satisfies the RNS-FHE criteria, this block completes the HoRNS-CNN model prediction process. To make inferences on the C_i , three cloud-deep CNN models were used.

These include the 42-layer Inception-V3 model, the 40-layer Two-Pathways Cascaded model, and the 152-layer ResNet-50 model. The hyperparameters of the models were chosen and tuned using the hyperband technique to provide accurate and efficient prediction, and their activations were approximated after training stability was achieved. Augmentation was also performed on cipher-images using the random erasing augmentation algorithm. The encrypted prediction outputs were subsequently transferred to the final block for decryption and performance analysis.

RNS homomorphic decryption block: This block utilizes the mixed-radix conversion (MRC) method as a reverse conversion algorithm for cipher-images. Specifically, C_i are broken into cipher-pixels c_i and their residues x_i w.r.t. moduli set m_i to produce an N -prime cipher-image form. Finally, the user will use the generated sk to retrieve the plain neuroimages from their corresponding cipher-images for performance analysis.

In summary, as illustrated in Fig. 1, the RNS Homomorphic Encryption Block comprises six distinct processes: pixel decomposition, key generation, random noise generation, moduli set computation, RNS forward conversion, and RNS bitstream encoding/encryption w.r.t. the defined special moduli set. Additionally, computations related to Homomorphic Privacy-Preserving Classification are performed in accordance with the computed moduli set to enhance energy efficiency. Finally, the RNS Homomorphic Decryption operations are executed w.r.t. the computed moduli set to ensure the overall energy efficiency of the system.

3.3 Formalization of RNS pixel-bitstream homomorphic encryption and decryption circuits

This section presents the mathematical modeling and design of the RNS pixel-bitstream homomorphic circuits for neuroimaging dataset encryption and decryption. These designs are presented w.r.t. given moduli set, m_i . Let us consider a 2D plain neuroimage $I_{i,j}$ of dimension $m \times n$ with pixels $X_{i,j}$ shown in the matrix of Eq. (14):

$$I_{i,j} = \begin{bmatrix} X_{1,1} & X_{1,2} & \cdots & X_{1,n-1} & X_{1,n} \\ X_{2,1} & X_{2,2} & \cdots & X_{2,n-1} & X_{2,n} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ X_{m-1,1} & X_{m-1,2} & \cdots & X_{m-1,n-1} & X_{m-1,n} \\ X_{m,1} & X_{m,2} & \cdots & X_{m,n-1} & X_{m,n} \end{bmatrix}, \quad (14)$$

where $1 \leq i \leq m$, $1 \leq j \leq n$ and $X_{i,j} \in \mathbb{Z}_n$ in the range $[0, 255]$ with 8-bit grayscale representation. The RNS-FHE algorithm returns a cipher-image $C_i = I_{i,j}'$ of the same dimension as shown in Eq. (15):

$$C_i = I_{i,j}' = \begin{bmatrix} X_{1,1'} & X_{1,2'} & \cdots & X_{1,n-1'} & X_{1,n'} \\ X_{2,1'} & X_{2,2'} & \cdots & X_{2,n-1'} & X_{2,n'} \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ X_{m-1,1'} & X_{m-1,2'} & \cdots & X_{m-1,n-1'} & X_{m-1,n'} \\ X_{m,1'} & X_{m,2'} & \cdots & X_{m,n-1'} & X_{m,n'} \end{bmatrix} \quad (15)$$

where each 8-bit length pixel $X_{i,j}$ has been encrypted to its corresponding cipher form $X_{i,j}'$ of equal bit-length. In other words, $X_{i,j}'$ is the concatenation of residue bits of pixel $X_{i,j}$ after encryption process. Assuming vector $\{x_1, x_2, \dots, x_n\}$ are the residues of $X_{i,j}$, then, the cipher-pixel, $X_{i,j}' = x_1 x_2 \dots x_n$. To achieve the above, there is the need to design a parallel n -bit sequence of RNS pixel-bitstream homomorphic encoder circuit using legitimate moduli set $\{m_1, m_2, \dots, m_n\}$ and a very strong key parameter ($\lambda = 4048$ -bit length) for the sk , where $n \geq 3$.

By using the given moduli set, $m_i = \{2^n - 1, 2^n, 2^{n+1} - 1\}$, where $m_1 = 2^n - 1$, $m_2 = 2^n$, and $m_3 = 2^{n+1} - 1$, we, first of all, propose the homomorphic encoding scheme for a neuroimage $I_{i,j}$. We begin the proposed design by establishing the legitimacy of the moduli set in Theorem 1 as follows:

Theorem 1: The moduli set $m_i = \{2^n - 1, 2^n, 2^{n+1} - 1\}$ is a legitimate set for the proposed RNS cryptosystem.

Proof: Given the moduli set $m_i = \{2^n - 1, 2^n, 2^{n+1} - 1\}$ where $m_1 = 2^n - 1$, $m_2 = 2^n$, and $m_3 = 2^{n+1} - 1$, we should be able to prove that the moduli are pairwise relatively-prime for any integer $n \in \mathbb{Z}_n$. By using the Euclidean theorem given in Eq. (16), we obtain the legitimacy of a pair of moduli as shown in Eq. (17):

$\gcd(a, b) = \gcd(b, |a|_b) = 1$. That is

$$\gcd(m_i, m_j) = \gcd(m_j, |m_i|_{m_j}) = 1 \text{ for } \forall i, j \text{ and } i \neq j, \quad (16)$$

Therefore, $\gcd(2^n - 1, 2^n) = \gcd(2^n, |2^n - 1|_{2^n}) = \gcd(2^n, -1) = 1$ also, $\gcd(2^n, 2^{n+1} - 1) = \gcd(2^{n+1} - 1, |2^n|_{2^{n+1} - 1}) = 1$ and finally,

$$\begin{aligned} & \gcd(2^n - 1, 2^{n+1} - 1) \\ &= \gcd(2^{n+1} - 1, |2^n - 1|_{2^{n+1} - 1}) = 1. \end{aligned} \quad (17)$$

Inference: Therefore, the given moduli set is legitimate and can be used to design the proposed RNS homomorphic cryptosystem.

3.3.1 Design of RNS pixel-bitstream homomorphic encoder circuit

The RNS pixel-bitstream homomorphic encoder can be designed according to [4, 54] as follows: To encode an 8-bit binary pixel X from a grayscale neuroimage, $I_{i,j}$ using moduli set $\{m_1, m_2, \dots, m_n\}$ with residue vector represented by $\{x_1, x_2, \dots, x_n\}$, the dynamic range (DR), M is computed using Eq. (18):

$$M = \prod_{i=1}^3 m_i = (2^n - 1) \cdot (2^n) \cdot (2^{n+1} - 1),$$

$$M = 2^{3n+1} - 2^{2n} + 2^n. \quad (18)$$

The expansion of M implies that pixel X is a $(3n + 1)$ -bit binary integer and sufficiently large enough for all $n \geq 3$, such that the expanded binary equivalent of X is of the form:

$$X = x_{3n} x_{3n-1} x_{3n-2} \dots x_1 x_0 \text{ for } \forall x_i \in \{0, 1\}. \quad (19)$$

The value of residues, x_i 's can be obtained by following the two Lemmas presented below.

Lemma 1: The i^{th} least significant bit (LSB) of X is derived simply from the modulo- 2^n RNS homomorphic processor as residue x_2 .

Lemma 2: For residue x_1 and x_3 respectively, the 8-bit integer value of pixel X is divided into three blocks comprising two n -bit blocks represented by Z_1 and Z_2 , and one $(n + 1)$ -bit block represented by Z_3 such that Eq. (20) holds.

$$\begin{aligned} Z_1 &= \sum_{j=2n}^{3n} x_j 2^{j-2n+1}, \\ Z_2 &= \sum_{j=n}^{2n-1} x_j 2^{j-n}, \\ Z_3 &= \sum_{j=0}^{n-1} x_j 2^j. \end{aligned} \quad (20)$$

Where Z_3 is the most significant bit (MSB). This implies that:

$$X = Z_1 + 2^n Z_2 + 2^{2n} Z_3. \quad (21)$$

From Lemma 2, x_2 which is the LSB is obtained directly from modulo- (2^n) RNS homomorphic processor as Z_1 . For x_1 and x_3 , two cases need to be considered following Lemma 2. These include:

Case 1: Designing modulo- $(2^n - 1)$ homomorphic processor. Designing this homomorphic modulo-processor produces x_1 as demonstrated in Eq. (22).

$$\begin{aligned} x_1 &= |X|_{2^n-1} = |Z_1 + 2^n Z_2 + 2^{2n} Z_3|_{2^n-1}, \\ x_1 &= \left| |Z_1|_{2^n-1} + |Z_2 2^n|_{2^n-1} + |Z_3 2^{2n}|_{2^n-1} \right|_{2^n-1}, \\ x_1 &= |Z_1 + Z_2 + Z_3|_{2^n-1}. \end{aligned} \quad (22)$$

Case 2: Designing modulo- $(2^{n+1} - 1)$ homomorphic processor. Like case 1, designing this homomorphic modulo-processor produces x_3 as demonstrated in Eq. (23).

$$\begin{aligned} x_3 &= |X|_{2^{n+1}-1} = |Z_1 + 2^n Z_2 + 2^{2n} Z_3|_{2^{n+1}-1}, \\ x_3 &= \left| |Z_1|_{2^{n+1}-1} + |Z_2 2^n|_{2^{n+1}-1} + |Z_3 2^{2n}|_{2^{n+1}-1} \right|_{2^{n+1}-1}, \\ x_3 &= |Z_1 + 2^n Z_2 + 2^{n-1} Z_3|_{2^{n+1}-1}. \end{aligned} \quad (23)$$

Example 1 illustrates the encoding process for pixel $X = 209$ using the above RNS pixel-bitstream homomorphic encoder circuit.

Example 1: Given a power-of-two moduli set $m_i = \{2^n - 1, 2^n, 2^{n+1} - 1\}$. if $n = 3$ and $X = 209$ (i.e., the decimal equivalent of an 8-bit integer), then, the encoding of X is as follows: Given that $X=209$, which is equivalent to an 8-bit integer 11010001. As illustrated above, $X = (3n + 1)$ -bit integer number. Since $n = 3$, we then divide X into two 3-bit blocks (Z_1 and Z_2) and a 2-bit block (Z_3) starting from the LSB as follows:

$Z_1=001$ (Decimal-1), $Z_2 = 010$ (Decimal-2), and $Z_3 = 11$ (Decimal-3).

From Lemma 1, $x_2 = Z_1 = 001 = 1$. The value of x_1 and x_3 can be computed using Eq. (22) and (23) according to Lemma 2 as follows:

$$x_1 = |Z_1 + Z_2 + Z_3|_{2^n-1} = |209|_{2^3-1} = |1 + 2 + 3|_7 = |6|_7 = 6(110),$$

$$x_3 = |Z_1 + 2^n Z_2 + 2^{n-1} Z_3|_{2^{n+1}-1} = |209|_{2^4-1} = |1 + 2^3(2) + 2^2(3)|_{15},$$

$$= |1 + 16 + 12|_{15} = |29|_{15} = 14(1110).$$

Summarily, the concatenation of residues $x_3 x_1 x_2 = 11101101$, which is equivalent to the decimal number 237. This implies that cipher-pixel X' will maintain the

same bit-length as plain-pixel X after encryption without bit overflow [4].

The hardware design of the RNS homomorphic encoder circuit is realized by constructing a circuit that comprises a parallel arrangement of n -number of combinations of four fast adders of two different types, as shown in Fig. 2a. Each combination is made up of two carry-save adders (CSAs), and two carry-propagate adders (CPAs). As illustrated in Fig. 2b, CSAs perform 3-bit additions, whereas CPAs perform 2-bit additions. Each CSA uses operands Z_1 , Z_2 , and Z_3 to generate partial sum (PS) and partial carry (PC), which are then fed into each CPA to generate residues. A CSA, unlike a ripple-carry adder (RCA), is made up of multiple combinations of 1-bit full adders (FA) with no carry chaining.

3.3.2 Design of homomorphic RNS pixel-bitstream decoder circuit

We employed the mixed radix conversion (MRC) algorithm to design a RNS pixel-bitstream homomorphic decoder circuit for cipher-pixels of a cipher-image I_{ij}' . Our choice is motivated by the inherent elimination of computationally expensive large modulo- M [55] that slows down the RNS reverse conversion process. However, the inter-dependency of mixed-radix digits (MRDs) makes the decoder circuit require more fast adders and computationally expensive than the designed homomorphic encoder circuit. Therefore, the decryption process requires more time to complete than the encryption process.

Given the residue of an 8-bit cipher-pixel $X' = (x_1, x_2, \dots, x_n)$ in an RNS with a set of relatively-prime integer moduli $\{m_1, m_2, \dots, m_n\}$, the 8-bit binary equivalent X can be computed using MRC algorithm defined in Eq. (24-25) as follows [55]:

$$X = a_1 + a_2 m_1 + a_3 m_1 m_2 + \dots + a_n m_1 m_2 m_3 \dots m_{n-1}, \quad (24)$$

where $a_{i=1, \dots, n}$, are the MRDs, each of which can be computed thus:

$$a_1 = x_1,$$

$$a_2 = \left| (x_2 - a_1) m_1^{-1} \right|_{m_2 m_3},$$

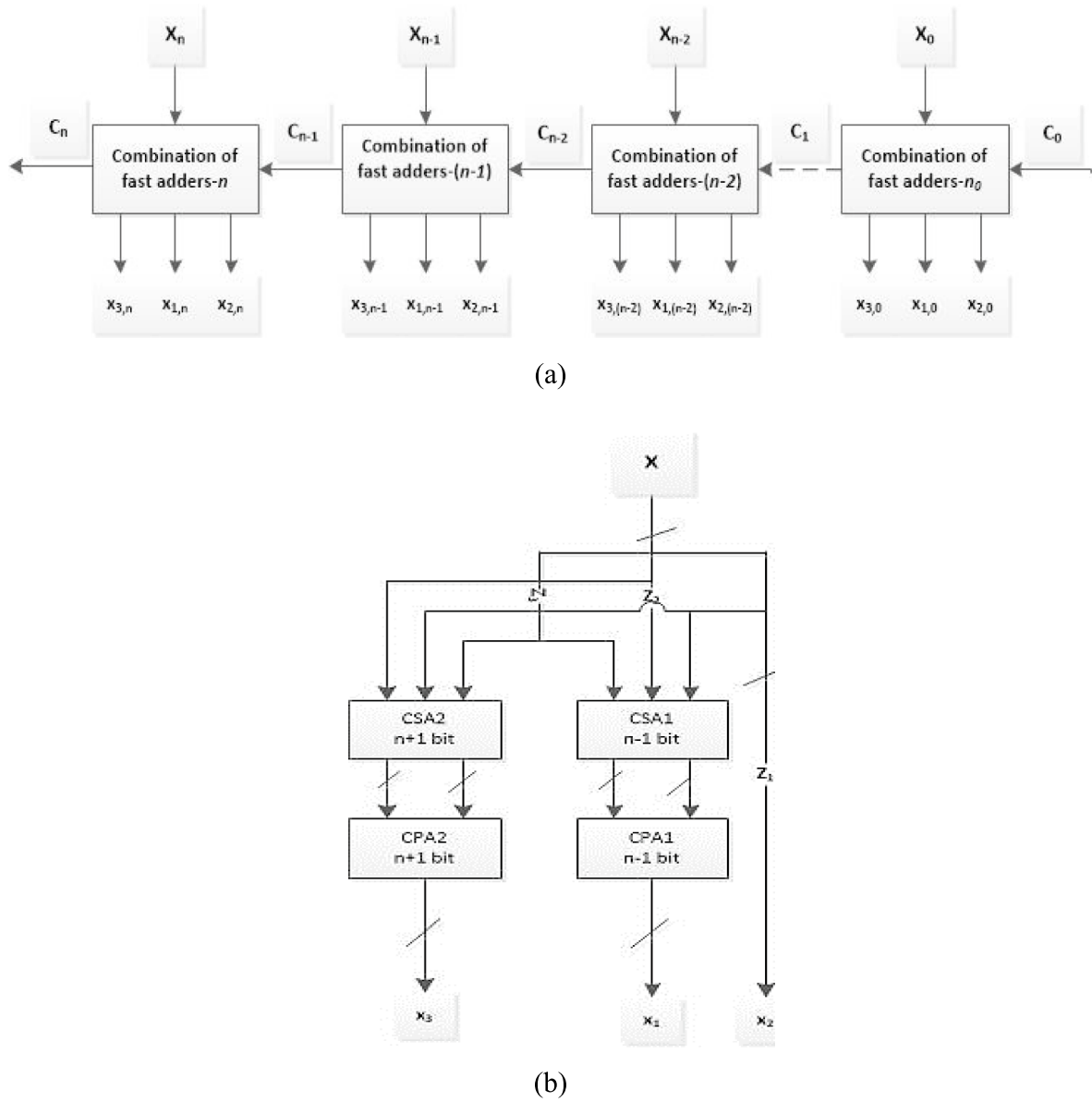


Fig. 2 Block diagram showing hardware realization of the proposed n -bit homomorphic RNS pixel-bitstream encoding circuit: **a** Parallel arrangement of n combinations of homomorphic encoder circuits **b** 1-bit circuit: CSA carry-save adder, CPA carry-propagate adder, PS partial sum, PC partial carry

$$\begin{aligned}
 a_3 &= \left| \left((x_3 - a_1) \left| m_1^{-1} \right|_{m_3} - a_2 \right) \left| m_3^{-1} \right|_{m_3} \right|_{m_3}, \\
 &\vdots \\
 a_n &= \left| \left(\dots \left((x_n - a_1) \left| m_1^{-1} \right|_{m_n} - a_2 \right) \left| m_{n-1}^{-1} \right|_{m_n} - \dots - a_{n-1} \right) \left| m_{n-1}^{-1} \right|_{m_n} \right|_{m_n}, \\
 &\quad (25)
 \end{aligned}$$

Since integer X is a pixel of binary digits from a gray-scale image I_{ij} , then, X has a unique 8-bit binary integer value in the interval $[0, M - 1]$, where the maximum value of pixel $X = 255(11111111)$.

Theorem 2: Given a vector of pairwise relatively-prime moduli set $m_i = \{m_1, m_2, \dots, m_n\}$ where $m_1 = 2^n - 1$, $m_2 = 2^n$, and $m_3 = 2^{n+1} - 1$ for every integer $n > 1$ in $\mathbb{Z}_m$ the following multiplicative inverses hold true:

$$\left| m_1^{-1} \right|_{m_2} = -1, \quad (26)$$

$$\left| m_2^{-1} \right|_{m_3} = 2, \quad (27)$$

$$\left| m_1^{-1} \right|_{m_3} = -2, \quad (28)$$

Proof: If it can be proved that $\left| m_i^{-1} \times m_i \right|_{m_i} = 1$, then m_i^{-1} is the multiplicative inverse of m_i w.r.t. m_i . The proof of multiplicative inverses is shown in Eqs. (26–28) are thus, demonstrated below:

From Eq. (26), we have.

$$\left| (2^n - 1) \times (-1) \right|_{2^n} = \left| (-1) \times (-1) \right|_{2^n} = 1,$$

From Eq. (27), we have.

$$\left| (2^n) \times 2 \right|_{2^{n+1}-1} = \left| 2^{n+1} \right|_{2^{n+1}-1} = \left| 1 \right|_{2^{n+1}-1} = 1,$$

Finally, for Eq. (28), we have.

$$\begin{aligned} & \left| (2^n - 1) \times (-2) \right|_{2^{n+1}-1} \\ &= \left| (-1) \times (2^{n+1} - 2) \right|_{2^{n+1}-1}, \\ &= \left| 2 - 1 \right|_{2^{n+1}-1} = \left| 1 \right|_{2^{n+1}-1} = 1. \end{aligned}$$

By substituting the values of multiplicative inverses given in Theorem 2 in Eq. (25), then, we have:

$$a_1 = x_1,$$

$$a_2 = \left| (x_2 - a_1)(-1) \right|_{2^n} = \left| x_2 + (-x_1) \right|_{2^n},$$

$$\begin{aligned} a_3 &= \left| ((x_3 - a_1)(-2) - a_2)(2) \right|_{2^{n+1}-1} = \left| 2^2 x_1 - 2^2 x_3 - 2a_2 \right|_{2^{n+1}-1}. \end{aligned} \quad (29)$$

And Eq. (24) becomes:

$$X = x_1 + 2^n a_2 - a_2 + 2^{2n} a_3 - 2^n a_3. \quad (30)$$

For hardware realization of the above homomorphic RNS decoder circuit, we provide further simplification of Eqs. (29–30) as follows: To start with Eq. (29),

$$a_1 = x_{1,n-1}x_{1,n-1} \dots x_{1,1}x_{1,0}, \quad (31)$$

$$\begin{aligned} a_2 &= \left| \underbrace{x_{2,n-1}x_{2,n-2} \dots x_{2,1}x_{2,0}}_{n\text{-bit}} + \underbrace{x_{1,n-1}x_{1,n-2} \dots x_{1,1}x_{1,0}}_{n\text{-bit}} \right|_{2^n} \\ a_2 &= \underbrace{a_{2,n-1}a_{2,n-2} \dots a_{2,1}a_{2,0}}_{n\text{-bit}} \end{aligned} \quad (32)$$

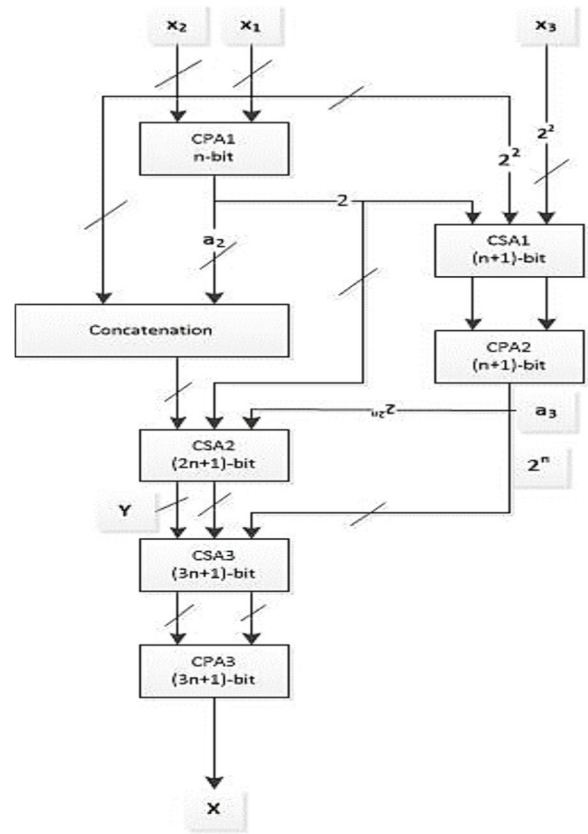


Fig. 3 Block diagram of the proposed 1-bit homomorphic RNS pixel-bitstream decoder circuit for pixel X

$$\begin{aligned} a_3 &= \left| \frac{2^2 (x_{1,n-1}x_{1,n-2} \dots x_{1,1}x_{1,0})}{n\text{-bit}} - \frac{2^2 (x_{3,n-1}x_{3,n-2} \dots x_{3,1}x_{3,0})}{(n+1)\text{-bit}} - \frac{2(a_{2,n-1}a_{2,n-2} \dots a_{2,1}a_{2,0})}{n\text{-bit}} \right|_{2^{n+1}-1}, \\ a_3 &= \underbrace{a_{3,n}a_{3,n-1} \dots a_{3,1}a_{3,0}}_{(n+1)\text{-bit}} \end{aligned} \quad (33)$$

Therefore, Eq. (30) can then be realized thus:

$$\begin{aligned} X &= \underbrace{Y_{2n-1}Y_{2n-2} \dots Y_0}_{2n\text{-bit}} + \underbrace{\hat{a}_{2,n-1}\hat{a}_{2,n-2} \dots \hat{a}_{2,0}}_{n\text{-bit}} + \\ & \underbrace{a_{3,n}a_{3,n-1} \dots a_{3,0}}_{(n+1)\text{-bit}} \underbrace{00 \dots 0}_{2n\text{-bit}} + \underbrace{\hat{a}_{3,n}\hat{a}_{3,n-1} \dots \hat{a}_{3,0}}_{(n+1)\text{-bit}} \underbrace{11 \dots 1}_{n\text{-bit}} \end{aligned} \quad (34)$$

$$X = \quad (35)$$

$$\text{where } Y = \underbrace{x_{1,n-1}x_{1,n-2} \dots x_{1,1}x_{1,0}}_{n\text{-bit}} \underbrace{00 \dots 0}_{n\text{-bit}} \underbrace{a_{2,n-1}a_{2,n-2} \dots a_{2,1}a_{2,0}}_{n\text{-bit}} \underbrace{00 \dots 0}_{n\text{-bit}}$$

and $\bowtie$ denotes a concatenation operator.

$$\begin{aligned} \text{That is, } Y &= \underbrace{x_{1,n-1}x_{1,n-2} \dots x_{1,1}x_{1,0}a_{2,n-1}a_{2,n-2} \dots a_{2,1}a_{2,0}}_{2n\text{-bit}} \\ &= Y_{2n-1}Y_{2n-2} \dots Y_1Y_0 \end{aligned} \quad (36)$$

The RNS pixel-bitstream homomorphic decoder circuit's block diagram contains an n -number of parallel combinations of fast adders with inverted structures (i.e., the inverted Fig. 2a). The circuit begins with the routing of the residues x_i , followed by the use of six simple fast adders: three CSAs and three CPAs to obtain the binary equivalent of the residues x_1 , x_2 , and x_3 . The architecture of each combination of the six fast adders is shown in the block diagram of Fig. 3.

3.4 Proof of homomorphic properties

Example 2: In Example 1, we demonstrated how a pixel with binary value 11010001 can be encrypted to 11101101 using the RNS pixel-bitstream homomorphic encoder circuit w.r.t. the given moduli set when $n = 3$. Using the same approach, another pixel with binary value 01111011 can be encrypted to 11100011. Homomorphic computations can be performed directly on the encrypted pixels or on the cipher-images. Let $X_1 = 11010001$ and $X_2 = 01111011$. Assuming the cipher form of X_1 is $C_1 = 11101101$ and the cipher form of X_2 is $C_2 = 11100011$ respectively and also, $sk = 00010111$ is the secret key. The illustration of the additive and multiplicative homomorphic properties is given below:

Additive Property:

$$C_3 = (C_1 + C_2) \bmod m_i$$

$$C_3 = (11101101 + 11100011) \bmod (00000111, 00001000, 00001111)$$

$$C_3 = 01001110$$

To decrypt C_3 , we apply

$$X_3 = [MRC(m_i, C_3)] \bmod MRC(m_i, sk)$$

$$X_3 = MRC\{([00000111, 00001000, 00001111], [01001110])\} \bmod$$

$$MRC([00000111, 00001000, 00001111], [00010111, 00010111, 00010111])$$

$$X_3 = 0101001100$$

$$\begin{array}{ccc} \text{In} & \text{other} & \text{words,} \\ X_3 = X_1 + X_2 = 11010001 + 01111011 = 0101001100 \end{array}$$

Multiplicative Property:

$$C_4 = (C_1 * C_2) \bmod m_i$$

$$C_4 = \{11101101 * 11100011\} \bmod \{00000111, 00001000, 00001111\}$$

$$\begin{aligned} C_4 &= \{0110100100010011\} \bmod \\ &\quad [00000111, 00001000, 00001111] \end{aligned}$$

$$C_4 = 0000010001111001$$

To decrypt C_4 , we apply

$$X_4 = [MRC(m_i, C_4)] \bmod MRC(m_i, sk)$$

$$X_4 = MRC$$

$$\{([00000111, 00001000, 00001111], [0000010001111001])\} \bmod$$

$$MRC([00000111, 00001000, 00001111], [00010111, 00010111, 00010111])$$

$$X_4 = 0110010001101011$$

$$\begin{array}{ccc} \text{In} & \text{other} & \text{words,} \\ X_4 = X_1 * X_2 = 11010001 * \\ 01111011 = 0110010001101011 \end{array}$$

3.5 Residue arithmetic units for homomorphic computations

The residue arithmetic units (RAU) for performing arbitrary homomorphic operations include modular adders, modular subtractors, and modular multipliers compatible with modulo of the form $(2^k \pm 1)$ where $k = n, n + 1$. The complexity of these arbitrary computations can be dealt with after decryption using the designed RNS pixel-bitstream homomorphic decoder circuit since these operations are partially dependent on this process.

- Addition:** Homomorphic addition operation can be performed using three modular adders—two for modulo- $(2^{n+1} - 1)$ and one for modulo- $(2^n - 1)$. Modulo- (2^n) adders have the simplest structure because they can be realized using an n -bit binary adder with ignored carry-out (C_{out}), thus, modulo- $(2^n - 1)$ and modulo- $(2^{n+1} - 1)$ adders receive considerable attention. When these adders are used to perform homomorphic additions, the size of the resulting cipher-image is the same as the size of the operands. Figure 4a depicts the proposed simplified structure of a modular adder for homomorphic additions.
- Subtraction:** A modulo- $(2^n - 1)$ subtractor can be realized by combining a modulo- $(2^n - 1)$ adder

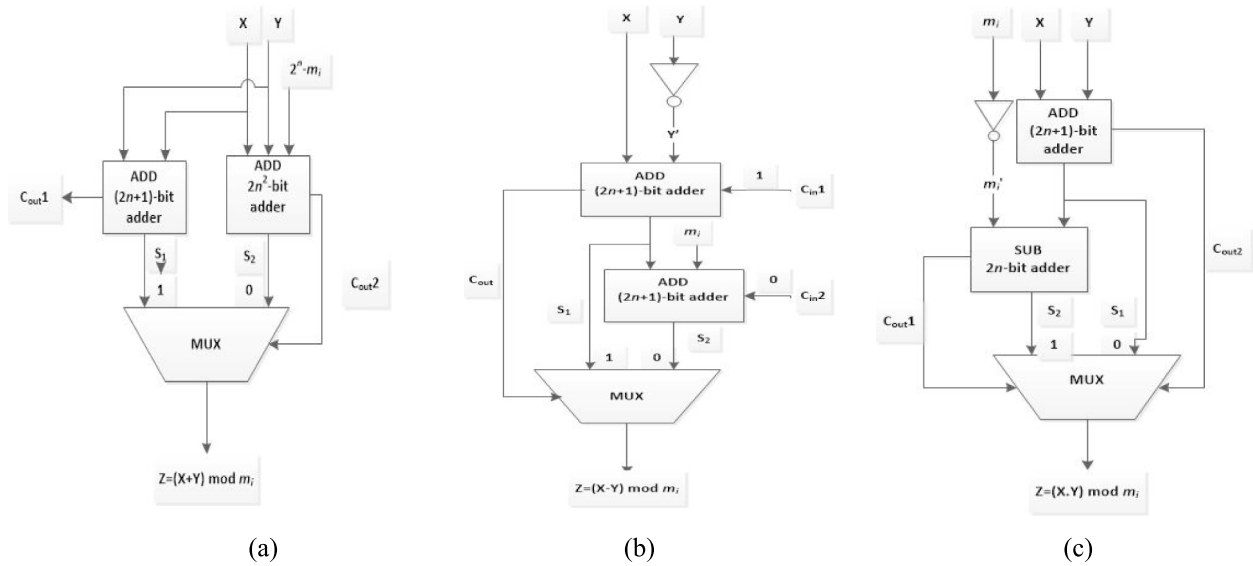


Fig. 4 Simplified RAU structure for homomorphic computations **a** RAU for homomorphic addition **b** RAU for homomorphic subtraction **c** RAU for homomorphic multiplication

and a few inverters (i.e., NOT gates) for homomorphic subtraction. Figure 4b shows the simplified structure of a modular subtractor setup for homomorphic subtractions.

- iii. **Multiplication:** The operation of homomorphic multiplication is very expensive in the sense that the size of the resulting cipher-image increases by up to l times (16-bit length in our example). In the worst-case scenario, the noise level increases significantly with polynomial asymptotic growth. This is due to the repeated additions that ensue between the operands with the significantly high noise level. In other words, the size of bits for representing each cipher-pixel overflows the valid range of $[0, 8)$ specified in the design. Redundancy can be used for overflow detection without correcting it. To keep the size of the resulting cipher-pixel in check within the acceptable 8-bit range without altering the result of the computation, there is, therefore, the need to perform a modular reduction after each multiplication operation. However, modular reduction with division is one of the most expensive computations in RNS. Studies have suggested the Montgomery reduction scheme to keep the size of homomorphic multiplications in check [33, 35]. Montgomery [35] proposed modular arithmetic modulo- N to keep multiplication products with sizes under control, where another modulus

$R > N$ is chosen so that modulus R and modulus N are pairwise relatively-prime. Divide by modulus R , intuitively, restores the operand size so that the modular reduction w.r.t. modulus N can be performed later. However, there is likely to be a trade-off between modular arithmetic computation efficiency and privacy in the current Montgomery scheme [33, 56]. In this study, we suggest modulo- $(2^k \pm 1)$ multiplier with $k = n$ and $k = n + 1$ respectively. This multiplier comprises two parts: the first part contains a comparison between modulo- $(2^n - 1)$ and modulo- $(2^{n+1} - 1)$, each of which belongs to different categories of modular multipliers. The second part depicts the proposed modulo- $(2^{n+1} - 1)$ multiplier, which employs the modulo- $(2^{n+1} - 1)$ subtractor as a fundamental component for modular reduction. Figure 4c depicts the simplified structure of a modular multiplier for homomorphic multiplications.

3.6 Simplified RNS-FHE algorithm for neuroimage data security

The proposed RNS-FHE scheme is a fully homomorphic encryption scheme that consists of three probabilistic polynomial time algorithms: key generation algorithm, encryption algorithm, and decryption as shown in Algorithm 3.3–3.5 below:

Algorithm 3.3 Key Generation Algorithm //This algorithm generates a pair of key parameters for RNS-FHE scheme encryption and decryption

Input: prime integers p, q
Output: Large prime integer N , Arbitrary secret key vector m_i^{-1} , Random noise, r_m
Procedure:
 1: KeyGen(p, q)
 2: Compute $N \leftarrow p * q$
 3: **for** $i=1$ to k
 4: choose relatively prime m_i such that dynamic range, $M = \sum_{i=1}^k m_i$
 5: **for** $j=1$ to k
 6: compute $m_i^{-1} = 1(\text{mod } m_i)$ such that $m_i^{-1} \in \mathbb{Z}_m$
 //where m_i^{-1} is an arbitrary vector of multiplicative inverse
 6: **return**(m_i^{-1})
 7: **end for**
 8: compute $r_m \leftarrow m_i^{-1} \cdot F_i(\text{rand})(\text{mod } m_i)$, for $i = (k + 1), \dots, (k + r)$
 // where $F_i(\text{rand})$ is a function to compute a vector of random numbers r
 9: **end for**
 10: **return**(N, r_m) //The values of N and r_m are the key parameters for encryption

Algorithm 3.4. Encryption Algorithm //This algorithm encrypts the plain neuroimages to cipher-images

Input: Neuroimage I , Parameter n , key
Output: Cipher-image C
Procedure:
 1: compute $rns_moduliset \leftarrow function(n)$
 2: **return**($z(2^n - 1, 2^n, 2^{n+1} - 1)$)
 3: compute $I_i \leftarrow X_1 X_2 \dots X_n$ //Neuroimage I_i is decomposed into their pixel pixels X_i
 4: $rns_pixel_encode \leftarrow function(X, n)$ //Pixel-bitstream encoding function
 5: $m = rns_moduliset(n)$
 6: **for** $i=1$ to $\text{length}(m)$
 7: $x[i] = X \text{ mod } m_i$ //RNS forward conversion process using Eq. (17)-(22)
 8: **return**(x)
 9: **end for**
 10: compute $c[i] \leftarrow Enc(x, key) = \{x * N \oplus key\} \text{ mod } m_i$ //where c is cipher-pixel
 11: compute $C \leftarrow concatenate(c)$
 12: **return**(C)

Algorithm 3.5 Decryption Algorithm**Input:** Cipher-image C , key, Parameter n **Output:** Neuroimage I **Procedure:**

```

1: compute  $key\_inverse \leftarrow solve(key)$  //This function compute the inverse of key parameters
2: for  $i = 1$  to  $length(m)$ 
3:   compute  $c^{-1}[i] \leftarrow \{C \oplus key\_inverse\} \bmod N$  //  $C$  is decomposed to  $c_i$ 
4:    $x[i] = c^{-1} \bmod m_i$ 
5:   return( $x$ )
6: end for
7:  $rns\_pixel\_decode \leftarrow function(x, n)$  //Pixel-bitstream decoding using Eq. (29)-(36)
8: compute  $X \leftarrow \{x * N \oplus key\_inverse\} \bmod m_i$ 
9: compute  $I \leftarrow concatenate(X)$ 
10: return( $I$ )

```

4 Experimental setup and simulation of the HoRNS-CNN model

Usman et al. [10] outline the specifications for the simulation of RNS pixel-bitstream homomorphic encoder and decoder circuits and for training the deep CNN networks used in the HoRNS-CNN model. The simulation experiments were carried out on a GPU system equipped with a 2.70GHz processor, 8.00GB of RAM, and a 4 Core(s) Intel (R) processor. After normalization, a patching algorithm [47] was used to patch each neuroimage data into 50 randomly overlapping 64×64 sizes to compensate for the large data requirements of DL training and reduce overfitting. The model was validated using tenfold cross-validation (CV), with 70% (331,520 patches) set aside for

training and 15% each for validation and testing (71,040 patches). Therefore, the overall train-test-validation split ratio for patches is 14:3:3. Patches containing dyslexia neural-biomarkers were assigned the binary digit 1, while control patches were assigned the binary digit 0. The architectures of deep CNNs used in the HoRNS-CNN model are described as follows:

- i. Inception-V3 CNN: The simulated Inception-V3 model has 42 layers and 11 Inception modules of five different types. Large filter sizes, such as 5×5 and 7×7 feature maps, were replaced with smaller asymmetric filters (1×7 and 1×5), and 1×1 convolution was used as a barrier before large filters

Table 2 Hyperparameters of the HoRNS-CNN model

Hyperparameters	Inception-V3	Cascaded model	ResNet50
Input image size	$117 \times 117 \times 3$	$117 \times 117 \times 3$	$224 \times 224 \times 3$
Input layer kernel size	3×3	3×3	3×3
Number of layers	42	40	152
First conv layer feature maps	$55 \times 55 \times 4$	$53 \times 53 \times 4$	$55 \times 55 \times 4$
First conv layer kernel size	5	3	5
First conv layer stride	3×3	3×3	3×3
Next few conv layer feature maps	$24 \times 24 \times 28$	$21 \times 21 \times 24$	$27 \times 27 \times 64$
No. fully connected layer (fc)	fc800	fc800	fc800
No. of parameters in million	23.2	22.8	25.6
Batch size	10,000 patches	10,000 patches	10,000 patches
Learning rate	0.001	0.001	0.001
Optimizer	SGD	SGD	SGD
Minimum training iterations	100 epochs	100 epochs	100 epochs
Maximum training iterations	550 epochs	550 epochs	550 epochs
Activation type	ReLU	ReLU	ReLU
Homomorphic activation type	Approximated ReLU	Approximated ReLU	Approximated ReLU
Decay value (constant)	0.0005	0.0005	0.0005
Momentum	0.9	0.9	0.9

to reduce the computational cost of this model. To deal with overfitting, the architecture was enhanced with dropout and batch normalization layers. These modules were concatenated in this model to achieve maximum extraction of the input data features [47].

- ii. Two-Pathway Cascaded CNN: The architecture of the Two-Pathway Cascaded feed-forward CNN model used to simulate the HoRNS-CNN model is similar to that of [4] with some minor modifications. The architecture keeps two CNN model pathways active. The input image is routed through two distinct convolution operation pathways. Con-

currently, the two pathways were trained to extract high-level features from cipher images. The first pathway is made up of smaller stacked receptive fields (7×7) and the second path is made up of larger 15×15 stacked receptive fields. After each convolution layer, the first CNN output line was connected directly to the first hidden layer of the second CNN model, and its outputs were concatenated with softmax activation. Fully-connected layers were used for training and classification, with a dropout layer included to avoid overfitting.

- iii. ResNet-50 CNN: The ResNet-50 network, which was used in the HoRNS-CNN model, has 152 lay-

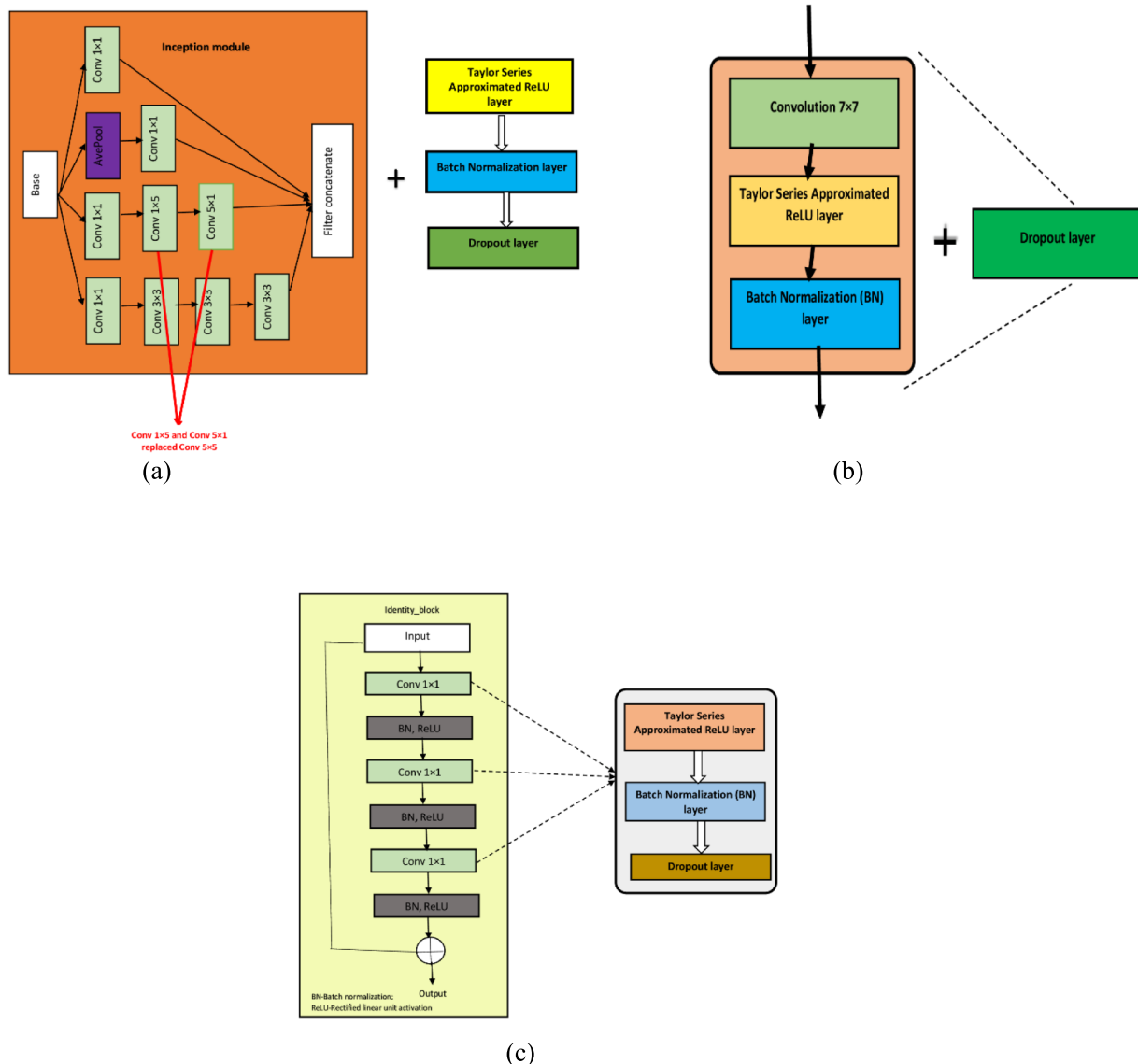


Fig. 5 Modifications to deep CNNs used in HoRNS-CNN model **a** Inception module of Inception-V3 **b** Two-Pathway Cascaded convolution layer **c** Identity block of ResNet-50

ers, 34 of which are plain and 34 of which are residual. The proposed architecture includes four stacks of ResNet blocks. Each ResNet block is made up of stacked layers of convolution blocks and repeated identity blocks, which represent two different types of shortcut modules. At the shortcut, the identity block lacks a convolutional layer. Consequently, its input retains uniform dimensions as its output. Because the convolutional layer is available at a shortcut, the input dimensions of a convolution block are smaller than the output dimensions. In both blocks, 1×1 convolution is implemented at the network's beginning and end using a technique known as bottleneck design [57]. This technique reduces the number of parameters while maintaining network performance. Finally, some of the deep

shortcut modules were replaced with a new set of classification layers tailored to the neuroimages being used.

The deep CNN models were first trained for stability, then adapted to homomorphic computations by approximating their ReLU activation function with a Taylor polynomial series of degree 3. To normalize the input feature maps within a certain range, a batch normalization layer was placed before each approximated activation layer. Training homomorphic CNN with higher-degree Taylor polynomial series, on the other hand, leads to a cipher-image expansion problem because the polynomial grows to infinity rather quickly, even for small x -axis values. Consequently, the outputs of an activation layer, as in the case of ResNet-50, are

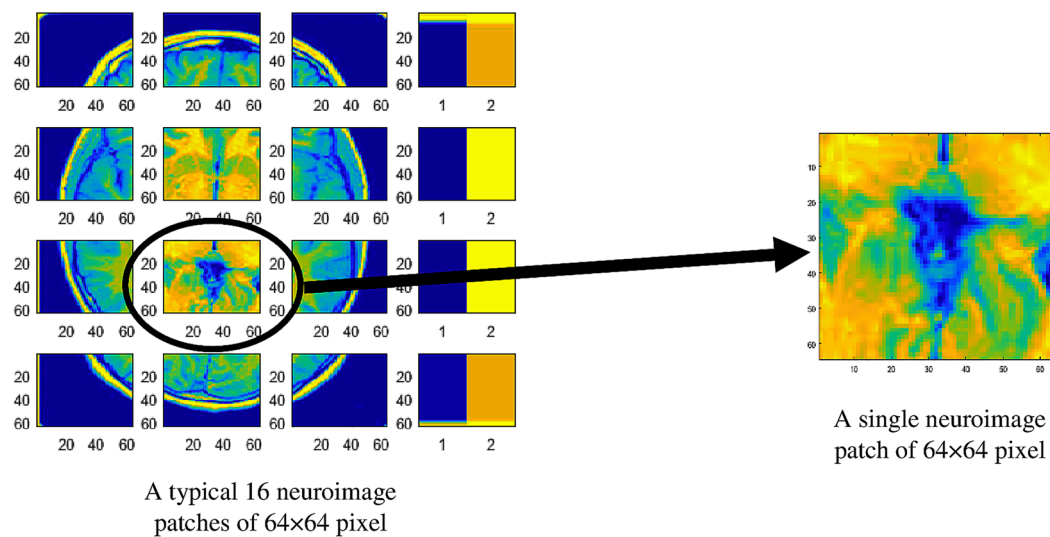


Fig. 6 An example of a typically patched neuroimage

Table 3 Critical path latency (Seconds) of homomorphic circuits (Virtex-4 FPGA)

<i>N</i>	Critical path	Proposed encoder	Liu et al. [59]	% improvement	Proposed decoder	Liu et al. [59]	% improvement
4	2^5-1	23.7	25.6	7.4	27.4	31.8	13.8
5	2^6-1	29.9	33.7	11.3	33.6	43.7	23.1*
8	2^9-1	37.1	48.5	23.5*	46.1	53.8	14.3
11	$2^{12}-1$	56.8	68.3	16.8	66.2	77.4	15.5

*Performance improvement is significantly higher than others

Table 4 Critical path latency (Seconds) of homomorphic circuits (Spartan-3 FPGA)

<i>n</i>	Critical path	Proposed encoder	Liu et al. [59]	% improvement	Proposed decoder	Liu et al. [59]	% improvement
4	2^5-1	25.5	30.1	15.3*	33.1	35.9	7.8
5	2^6-1	35.2	39.7	11.2	41.7	52.8	21.0*

*Performance improvement is significantly higher than others

extremely large, resulting in a cipher expansion [8, 10, 58]. Table 2 presents a summary of the hyperparameters adjusted during the training process. The hyperparameters of the CNN models and the homomorphic operations in the proposed HoRNS-CNN model were optimized using the Hyperband optimization method. The choice of Hyperband was essential due to the significant computational complexity associated with FHE operations, particularly addition and multiplication, which require extensive computational resources. Compared to traditional hyperparameter optimization techniques such as random search and Bayesian optimization, Hyperband utilizes an adaptive resource allocation strategy that efficiently distributes computational resources to high-performing configurations while systematically discarding suboptimal ones. This approach improves optimization efficiency and minimizes computational overhead. Figure 5 and 6 depict the modifications made to certain layers of deep CNN models and typical patched neuroimage data.

5 Results and discussion

5.1 Evaluation of energy efficiency of the proposed RNS-FHE scheme

Energy efficiency of the RNS-FHE scheme is evaluated based on three metrics used to evaluate the performance of RNS pixel-bitstream encoder-decoder homomorphic circuits. These include critical path delay, area-square complexity, and homomorphic circuit latency.

5.2 Critical path latency or delay

To evaluate the energy efficiency of the RNS-FHE scheme, the critical path latency or delay (in seconds) of the simulated RNS pixel-bitstream homomorphic encoder-decoder circuits was measured and compared

to the state-of-the-art RNS scheme proposed by Liu et al. [59]. Tables 3 and 4 show the computed critical path delay during simulation with two different FPGAs. A critical path is a channel with the maximum latency as a result of the evolved high complexity of homomorphic computations and hardware requirements. The critical channel of the used moduli set is modulo- $(2^{n+1} - 1)$, where the largest amount of homomorphic computational time is required.

The homomorphic circuits were first of all, simulated on a pure adder-based Virtex-4 XC4VSX25 FPGA for different critical path values: 2^5-1 , 2^6-1 , 2^9-1 , and $2^{12}-1$ bits. When $n=4, 5, 8$, and 11 , the proposed design's timing efficiency was outstanding, with maximum frequencies of 353.4MHz, 292.8MHz, 275.8MHz, and 231.3MHz, respectively. Thereafter, the Spartan-3 XC3S200 FPGA was used for the same simulations. However, due to a lack of integrated block random access memory (BRAM) count, the circuits could only be implemented with a maximum frequency of 383.4MHz and 258.1MHz for two critical paths (2^5-1 and 2^6-1 bits) on this FPGA, when $n=4$ and 5 . Meanwhile, an unexpected finding was that the critical path latency of the proposed designs on FPGAs, for $n=3$ (2^4-1 bits), was observed to be 42.4% lower than that of the ROM-based simulation. This result indicates that the FPGA implementation achieves a 42.4% improvement in energy-efficiency compared to the ROM-based simulation. When $n=8$, the proposed homomorphic encoder circuit improves performance by 23.5% on Virtex-4 FPGA, while the homomorphic decoder circuit improves performance by 23.1% when $n=5$. The best performance improvement for the homomorphic encoder circuit on Spartan-3 FPGA is 15.3% when $n=4$, while the best performance improvement for

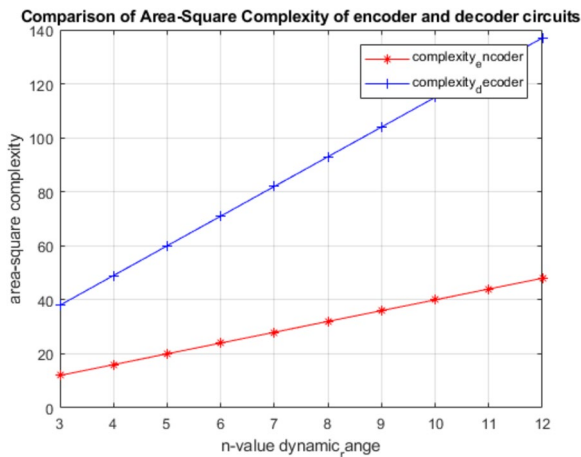


Fig. 7 Comparison of area-square complexity of homomorphic circuits

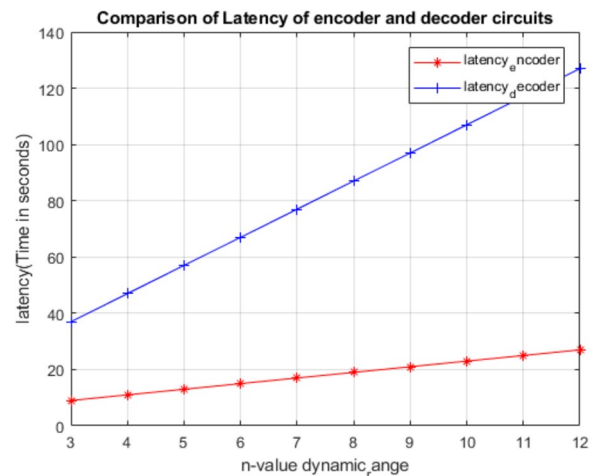


Fig. 8 Comparison of Latency (in seconds) of homomorphic circuits

the homomorphic decoder circuit is 21.0% when $n=5$ on the same circuit board. According to the two tables, the critical path latency can only be simulated on Spartan-3 FPGA for the peak of $n=8$. Due to the computational complexity of the modulo- $(2^{n+1} - 1)$ and the lack of integrated BRAM count. Attempting to simulate them on FPGAs will be ineffective for applications that require a larger n value. External ROMs are not advised because they are significantly slower than built-in ROMs. Therefore, for applications requiring a small value of n , such as digital image processing, the ROM-based platform is adequate and can outperform adder-based platforms. The proposed homomorphic circuits, on the other hand, is energy-efficient and preferable for applications that require a larger value for n .

5.3 Area-square complexity

Figure 7 depicts the comparison of the area-square complexity of the homomorphic encoder and decoder circuits as the dynamic range value n increases. From this figure, it can be deduced that the homomorphic encoder circuit requires a significantly lesser number of fast adders for its design than the homomorphic decoder circuit as the value of n increases. All the computed area-square complexity results were determined to be statistically significant at the 95% confidence interval (CI) level. The above CI level was established by calculating the mean difference and standard deviation following a tenfold cross-validation (CV) procedure. Thus, the depth of homomorphic computations performed by the homomorphic encoder is significantly reduced resulting in low time complexity and less energy consumption. The implication of this is that the homomorphic RNS pixel-bitstream encoder circuit will complete several batches of neuroimage patches encryption in a linear asymptotic time complexity of $O(n)$. Evidence supporting the deduction is the computation of area-square complexities (A) of each n -bit circuit as shown below.

$$A_{encoder} = [2(n+1) + 2(n-1)]A_{FA},$$

$$A_{encoder} = 4nA_{FA},$$

$$A_{decoder} = [n + 2(n+1) + (2n+1) + 2(3n+1)]A_{FA},$$

$$A_{decoder} = (11n+5)A_{FA}.$$

5.4 Homomorphic circuit latency

Figure 8 shows the estimation of the homomorphic circuits' latency (in seconds). Also, the homomorphic decoder circuit requires more time than the homomorphic encoder circuit. All the latency results obtained

were found to be statistically significant at the 95% CI level. This was determined by computing the mean difference and standard deviation after applying a tenfold CV technique. The building block of the homomorphic RNS pixel-bitstream encoder-decoder circuits are mainly fast adders (CPAs and CSAs) as shown in Figs. 2, 3 and 4. The CPA requires an area-square complexity similar to CSA but possesses twice the latency of CSA if there is no carry-in (C_{in}) of bits. The homomorphic circuit latency (D) that was imposed on homomorphic encryption and decryption processes for each n -bit circuit is computed below:

$$D_{encoder} = [(n+1) + 2(n+1)]D_{FA},$$

$$D_{encoder} = (3n+3)D_{FA}.$$

$$D_{decoder} = [2n + 2(n+1) + 2(3n+1) + 3]D_{FA},$$

$$D_{decoder} = (10n+7)D_{FA}.$$

5.5 Evaluation of asymptotic efficiency of the RNS-FHE scheme

To evaluate the asymptotic efficiency of the RNS-FHE scheme, the multiplication operation was considered due to its high energy consumption and greater computational complexity compared to the homomorphic addition and subtraction operations. During homomorphic multiplications, the proposed scheme spends most of its time in the worst-case scenario due to the depth of repeated additions in conjunction with the high level of noise that is involved in the operation. The time-complexity of the scheme is computed by considering the product of two sets of vectors $\{C_{ij}, C^f\}$ with radices $\{C_{ij}^b\}$ to represent entries of cipher-pixel c_{ij} and c_{rj} as product $c_{ij} \times C_{ij}$ and $c_{oj} \times C^r$ respectively for $i = 1, \dots, l$, $j = 1, \dots, k$, $o = 1, \dots, r$. Based on this product, however, some rules were defined as follows:

Rule 1:

$$C_{ia} \times C_{ib} = \sum_{\alpha=1}^k \sum_{\beta=1}^l \gamma_{iajb}^{\alpha\beta} \cdot C_{\alpha\beta} + \sum_{s=1}^r \gamma_{iajb}^s \cdot C^s,$$

$$\text{Rule 2 : } C_{ia} \times C^f = \sum_{\alpha=1}^k \sum_{\beta=1}^l \gamma_{fa}^{g\alpha\beta} \cdot C_{\alpha\beta} + \sum_{s=1}^r \gamma_{fa}^{gs} \cdot C^s \quad (37)$$

Rule 3:

$$C^f \times C^g = \sum_{\alpha=1}^k \sum_{\beta=1}^l \gamma_{fg\alpha\beta} \cdot C_{\alpha\beta} + \sum_{s=1}^r \gamma_{fgs} \cdot C^s,$$

where γ is the system's pk estimated as:

$$\gamma_{iajb} = \left(C_{ia}^b \cdot S^{-1}(\text{mod} m_i) \right) \times \left(C_{ja}^b \cdot S^{-1}(\text{mod} m_i) \right), \quad (38)$$

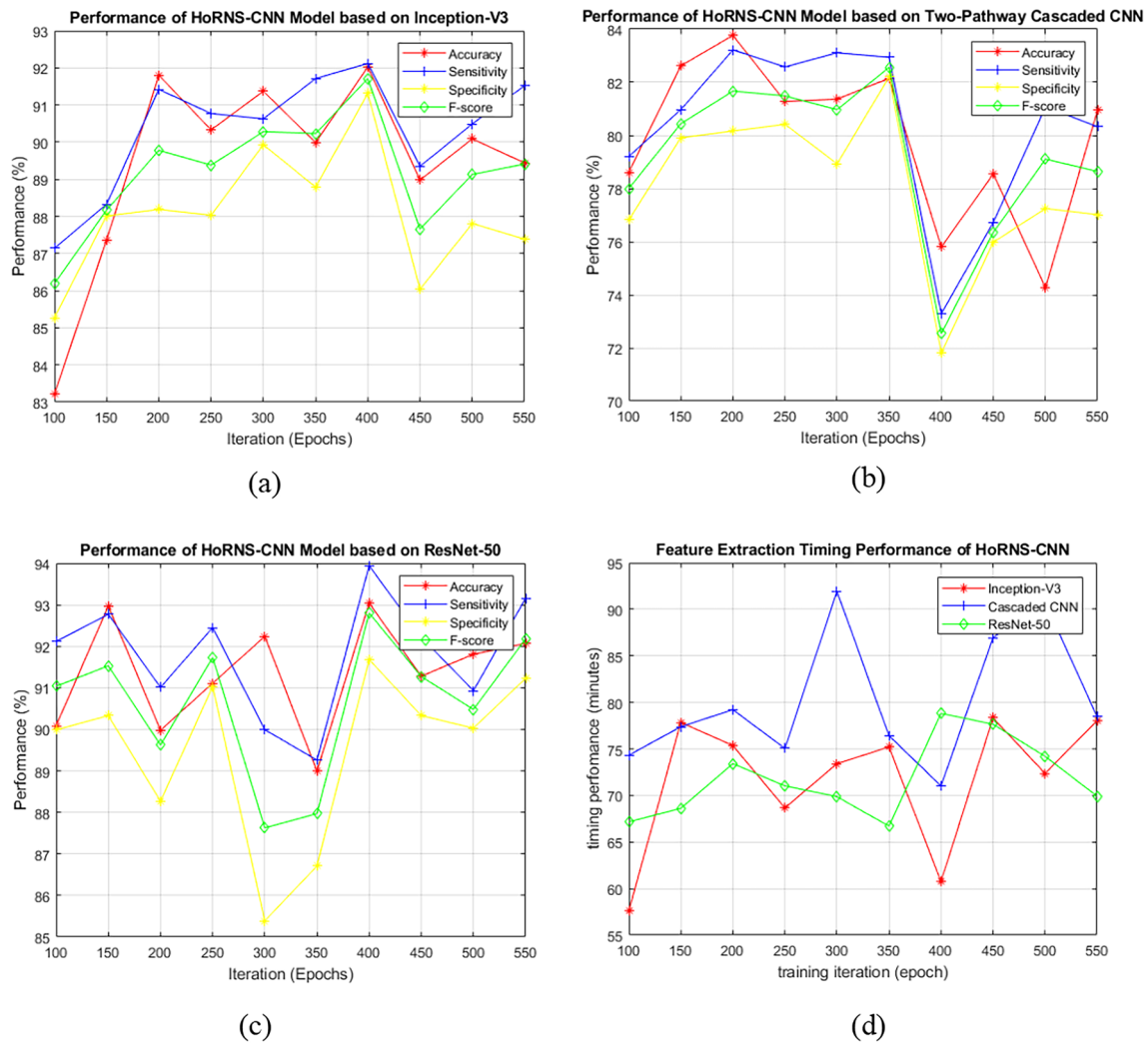


Fig. 9 HoRNS-CNN model classification performance: **a** Inception-V3 model **b** Two-Pathway Cascaded CNN model **c** ResNet-50 model

Table 5 Classification performance of HoRNS-CNN model on cipher neuroimages (mean $\pm$ SD)

HoRNS-CNN Model	Iteration (Epoch)	Accuracy (%)	Sensitivity (%)	Specificity (%)	F-Score (%)	Performance
Inception-V3	100	83.21 $\pm$ 0.0185	87.14 $\pm$ 0.0344	85.27 $\pm$ 0.0282	86.19 $\pm$ 0.0374	Worst
	400	92.02 $\pm$ 0.0176	92.12 $\pm$ 0.0219	91.31 $\pm$ 0.0187	91.71 $\pm$ 0.0352	Best
		89.46 $\pm$ 0.0244	90.35 $\pm$ 0.0237	88.08 $\pm$ 0.0248	89.19 $\pm$ 0.0276	Average
Two Pathway Cascaded CNN	500	74.26 $\pm$ 0.0391	81.07 $\pm$ 0.0267	77.25 $\pm$ 0.0199	79.11 $\pm$ 0.0278	Worse
	200	83.77 $\pm$ 0.0087	83.22 $\pm$ 0.0221	80.17 $\pm$ 0.0212	81.67 $\pm$ 0.0133	Best
		79.94 $\pm$ 0.0244	80.35 $\pm$ 0.0236	78.05 $\pm$ 0.0415	79.18 $\pm$ 0.0241	Average
ResNet-50	350	88.99 $\pm$ 0.0346	89.26 $\pm$ 0.0318	86.71 $\pm$ 0.0304	87.97 $\pm$ 0.0264	Worst
	400	93.04 $\pm$ 0.0300	93.94 $\pm$ 0.0350	91.68 $\pm$ 0.0255	92.80 $\pm$ 0.0235	Best
		91.36 $\pm$ 0.0252	91.79 $\pm$ 0.0223	89.50 $\pm$ 0.0199	90.63 $\pm$ 0.0222	Average

Bold means statistically significantly larger (p-value 0.05), 95% CI level for classification metrics

From the above rules, it can be deduced that the size of the resulting homomorphic multiplication is equal to $(k.l + r)$. As the user of homomorphic multiplication results sends a multiplication look-up table with pre-computed values of γ elements to the HoRNS-CNN model in the cloud, the model is left to perform $(k.l + r)$ multiplications for asymptotic complexity of $O((k.l)^2 + r.k.l)$ units of cipher images. Therefore, the asymptotic complexity of the proposed RNS-FHE scheme is thus, defined as:

$$O(k.l(k.l + r)^2) = O((k.l)^3 r) = O(n^3). \quad (39)$$

In summary, the RNS-FHE scheme requires probabilistic polynomial time growth of $O(n^3)$ to perform homomorphic computations, thus corroborating the concept of the FHE scheme.

5.6 HoRNS-CNN model classification performance

Figure 9 depicts the classification performance of the HoRNS-CNN model on cipher neuroimage (cipher-image). The results of each pre-trained deep CNN model were obtained experimentally using standard ML metrics such as accuracy, sensitivity, specificity, and F-Score. For the best-trained model, AUC values were obtained. Each result was calculated using 10 repeated tenfold CV at 95% CI (p -value < 0.05) at different training iterations (100–550 epoch). The high-level feature extraction time of each model was calculated during the prediction of encrypted neural-biomarker features. Table 5 summarizes the worst, best, and average performance, respectively.

Table 6 Comparison of ResNet-50 AUC values on plain and cipher neuroimages

Iteration (epoch)	Plain neuroimages AUC	Cipher neuroimages AUC
100	0.8377 ± 0.0192	0.7295 ± 0.0123
150	0.8415 ± 0.0391	0.6892 ± 0.0195
200	0.8465 ± 0.0307	0.7611 ± 0.0167
250	0.8308 ± 0.0142	0.7744 ± 0.0153
300	0.8580 ± 0.0239	0.7807 ± 0.0083
350	0.8701 ± 0.0144	0.7805 ± 0.0278
400	0.8795 ± 0.0256	0.7797 ± 0.0350
450	0.8911 ± 0.0391	0.7808 ± 0.0277
500	0.8897 ± 0.0185	0.7702 ± 0.0135
550	0.8972 ± 0.0226	0.7844 ± 0.0178

Table 5 and Fig. 9 show that deep CNN models in HoRN-CNN used an energy efficient pixel-based homomorphic privacy-preserving mechanism and could perform arbitrary homomorphic computations directly on cipher-images within the framework of the RNS-FHE scheme. They also allow for data augmentation on cipher-images [30]. These models achieved cutting-edge classification performance while requiring less future extraction time.

As shown in Fig. 9a, the accuracy of the Inception-V3 model ranges from 83.21% to 92.02% with an average of 89.46%, the sensitivity ranges from 87.14% to 92.12% with an average of 90.35%, the specificity ranges from 85.27% to 91.31% with an average of 88.08%, and the F-score ranges from 86.19% to 91.71% with an average of 89.19%. The best testing performance was obtained with 400 epochs in 61 min of feature extraction time. This model takes an average of 73 min to extract high-level features from cipher-images. As illustrated in Fig. 9b, the accuracy of the Two-Pathway Cascaded CNN model ranges from 74.26% to 83.77% with an average of 79.94%, the sensitivity ranges from 73.29% to 83.22% with an average of 80.35%, the specificity ranges from 71.81% to 82.21% with an average of 78.05%, and the F-score ranges from 72.54% to 82.57% with an average of 79.18%. The best testing performance was obtained with 200 epochs in 79 min of feature extraction time. This model takes an average of 80 min to extract high-level features from cipher-images. Finally, the accuracy of the ResNet-50 model ranges from 88.99% to 93.04% with an average of 91.36%, the sensitivity ranges from 89.26% to 93.94% with an average of 91.79%, the specificity ranges from 86.71% to 91.68% with an average of 89.50%, and the F-score ranges from 87.63% to 92.80% with an average of 90.63%, as shown in Fig. 9c. The best testing performance was achieved with 400 epochs and 79 min of feature extraction time. This model extracts high-level features from cipher-images in an average of 72 min. The ResNet-50 model has a significantly shorter feature extraction time than the Inception-V3 and Two-Pathway Cascaded CNN models, as shown in Fig. 9d.

By implication, Inception-V3 correctly predicts 423,680 cipher-image patches in approximately 73 min, the Two-Pathway Cascaded CNN model correctly predicts 378,596 cipher-image patches in approximately 80 min, and the ResNet-50 model correctly predicts 432,681 cipher-image patches in approximately 72 min. The ResNet-50 model outperforms the other two deep CNN models in terms of privacy-preserving classification performance of neural-biomarkers features. Table 6 summarizes the AUC values obtained from the ResNet-50 model on both plain and cipher-images. This table shows

Table 7 Performance comparison of HoRNS-CNN model with baseline

Authors	Dataset	Model name	HE algorithm type	Accuracy (%)	Reference No.
Tanaka	CIFAR-10	Pyramidal Residue Network	Block-based	56.8	[25]
Sitaula et al.	Flickr images	–	Block-based	85.69	[66]
Sirichotedumrong et al.	CIFAR-10	ResNet-18	Pixel-based	86.99	[30]
Chao et al.	Retinopathy of Prematurity (ROP) images	CaRENets	Pixel-based	92.50	[15]
	Diabetic Retinopathy (DR) image			72.50	
Proposed	MRI datasets	HoRNS-CNN	Pixel-based	91.36*	

that the AUC values obtained with plain neuroimages are slightly greater than those obtained with cipher-images.

5.7 Discussion of findings and performance comparison

The findings of this study confirmed that neuroimaging datasets contain some essential neural-biomarker features that can be used to discriminate brain-related neurodevelopmental disorders using advanced ML techniques. These characteristics are composed of grey matter (GM), white matter (WM), and cerebrospinal fluid (CSF) tissues in specific regions of the brain that house the phonological, cognitive, and auditory subsystems [53, 60]. Based on the above theoretical foundation, our method is consistent with those of [60–64], with the addition of the security element.

The findings also confirmed that an efficient HE privacy-preserving classification method can be designed for neural-biomarkers using a secured energy-efficient RNS-FHE scheme and deep CNN models. Therefore, the proposed HoRNS-CNN model extracts high-level neural-biomarker features with state-of-the-art classification performance. The study illustrated the RNS-FHE scheme's homomorphism properties and found that the scheme is homomorphic w.r.t arbitrary addition, subtraction, and multiplication operations when moduli set $\{2^n - 1, 2^n, 2^{n+1} - 1\}$ was used and the value of $n \geq 3$. The proposed scheme also has a manageable cipher-image expansion property with a polynomial asymptotic time-complexity of $O(n^3)$ and is secured against some known attacks (e.g., brute-force) as illustrated in Usman et al. [8]. When compared to the scheme by Liu et al. [59], the FPGA simulation of the RNS-FHE scheme improves critical path latency by 23.5% and 15.3% for both encoding and decoding homomorphic circuits. The simulations also show that the scheme requires fewer fast adders without the use of external ROMs, saving 42% more energy than its equivalent [59]. The performance

of pre-trained CNN models used in the HoRNS-CNN model achieved a significantly high prediction accuracy on cipher images.

Table 7 compares the proposed HoRNS-CNN model's performance to recent studies based on HE privacy-preserving classification. As shown in this table, our method outperforms state-of-the-art methods, demonstrating the feasibility of classifying low-resolution grayscale medical images [15] with reasonable feature extraction time. All deep CNN models used in HoRNS-CNN outperform Tanaka's [25] baseline prediction accuracy of 50%. This was observed with the cipher neural-biomarker features in the neuroimages used. Dowlin et al. [13] used a CNN with two convolutional layers and two fully connected layers to construct CryptoNets for the privacy-preserving classification of MNIST and CIFAR-10 image datasets. The network uses the Sigmoid activation function and replaces the max-pooling layer with scaled average-pooling and the ReLU function with the square function (x^2). Although, CryptoNets achieved 98.95% accuracy on the cipher-images, deeper CNN models could not be implemented using this network due to a depth limitation imposed by an unbounded derivative on homomorphic operations that characterized square function (x^2), which was used to approximate the network's activation functions. The table further demonstrates that the proposed HoRNS-CNN model outperforms the state-of-the-art equivalents by Chabanne et al. [18] and Rouhani et al. [65]. These two methods improved networks performance by using the Taylor series to approximate the ReLU function and batch normalization layers before each activation, but with higher-degree polynomials ($k \geq 4$). Although we used the same Taylor series to approximate ReLU activation in the pre-trained CNN models, however, we discovered that implementing with polynomial order greater than degree-3 resulted in significantly larger cipher-expansion. This is due to the depth of these networks. To address this issue, we kept

polynomial order of degree-3 with the ability to augment the cipher-images in order to manage cipher-expansion during homomorphic computations.

Tanaka's [25] method had the lowest classification accuracy of 56.8% because of the type of image encryption algorithm used. The adaptation layer block of the classifier overlaps with the adjacent cipher block. The HoRNS-CNN method keeps the location of pixel collections representing ROIs of important features and neural-biomarkers in the cipher-image even after the RNS-FHE scheme modifies the bitstream content of the pixel. This allows us to achieve state-of-the-art accuracy for all simulated deep CNN models. Furthermore, due to the computational overhead of the higher-order polynomial activation function, which limits the depth of homomorphic multiplications and thus, extremely large cipher-expansion, our proposed method outperforms the Chao et al. [15] and Sitaula et al. [66] methods. Finally, the proposed HoRNS-CNN model performed 1.14% worse than that of [15] due to a large number of CNN hyperparameters that were adjusted during the training session. Therefore, selecting an appropriate data augmentation and optimizing hyperparameters remains an open challenge in DL data analysis. When compared against the equivalent on plain neuroimages [47], the performance of the HoRNS-CNN model depreciated by 3.3% with a long feature extraction time. Therefore, future research will concentrate on developing a holistic method of dealing with hyperparameters optimization and cipher-image augmentation. When compared to the current state-of-the-art method, the deep CNN models in the HoRNS-CNN model achieve significantly higher classification performance across all metrics.

While many existing privacy-preserving methods provide strong security, they often do so at the expense of efficiency and classification accuracy. For example, models such as CryptoNets, FHE-CNNs, and leveled-FHE frameworks facilitate encrypted inference while safeguarding patient data. However, these approaches incur significant computational overhead due to complex polynomial arithmetic, modular transformations, and large ciphertext sizes. In contrast, the proposed HoRNS-CNN model optimizes FHE parameters using a RNS to enhance arithmetic operations, thereby reducing latency and energy consumption as shown in the preceding sections while preserving privacy. Additionally, differential privacy (DP)-based CNN models, such as DP-SGD, introduce controlled noise into training datasets to prevent individual re-identification. While these methods ensure robust privacy protection, they often degrade classification accuracy due to noise injection. Unlike DP approaches, the proposed HoRNS-CNN model does not rely on noise perturbation, thereby achieving high

classification accuracy while maintaining full encryption of patient data. In summary, the HoRNS-CNN model differentiates itself by integrating an optimized RNS-FHE scheme with deep CNNs, ensuring the privacy preservation of neural-biomarker features while minimizing computational and energy overhead. This balance positions HoRNS-CNN model as a promising solution for privacy-preserving neural-biomarker classification in medical applications.

6 Conclusion and practical implications

The HoRNS-CNN model was presented in this study for the privacy-preserving classification of dyslexia neural-biomarkers from publicly available neuroimaging datasets. The HoRNS-CNN model is primarily composed of an energy-efficient RNS-FHE scheme and pre-trained deep CNN models. Specifically, we provide a mathematical formalization of algorithms for designing the RNS pixel-bitstream homomorphic encoder-decoder circuits, as well as an encryption/decryption framework for neural-biomarker features inherent in the used neuroimages, as a follow-up to [4, 10] while the security analysis of the proposed scheme have been elaborated in [8]. Their cipher-images were subjected to remote DL classification, also known as the privacy-preserving prediction. We were able to accomplish the above task by adjusting the hyperparameters of deep CNN models and using a powers-of-two special moduli set $\{2^n - 1, 2^n, 2^{n+1} - 1\}$. We demonstrated the RNS-FHE scheme's homomorphism properties and provided some proofs of our moduli set legitimacy as well as proof to ensure the scheme's security. The following are the contributions of this work:

1. The proposed energy-efficient RNS-FHE scheme and homomorphic circuits are pure adder-based cryptosystems that can encrypt 8-bit grayscale normalized neuroimages while preserving pixel locations that contain relevant neural-biomarker features.
2. The homomorphic circuits' FPGA simulation indicates that they could save up to 42.4% energy when compared to their ROM-based equivalent. This results in a 23.5% and a 23.1% reduction in critical path latency for both the homomorphic encoder and the decoder circuits, respectively. However, because of the computational complexity of homomorphic multiplication on cipher images, the decryption process has a polynomial asymptotic time-complexity of $O(n^3)$.
3. The deep CNN algorithms used in the HoRNS-CNN model are efficient in terms of asymptotic complexity, energy-saving, feature extraction timing per-

formance, and classification metrics used without revealing the dataset's secret.

4. The Inception-V3 model predicts 423,683 cipher-image patches in 73 min and has an average accuracy of 89.5%. The Two-Pathway Cascaded CNN model predicts 378,596 cipher-image patches in 80 min and has an average accuracy of 79.9%. The ResNet-50 model predicts 432,681 cipher-image patches in 72 min and has an average accuracy of 91.4%.

In recent years, DL models have attracted significant attention in the healthcare sector due to state-of-the-art advancements in various fields. Despite their potential for offering personalized medical solutions, stringent regulations on patient health information confidentiality have often impeded the integration of DL-based solutions into clinical workflows. To overcome this issue and facilitate the processing of sensitive health information without compromising data privacy, we propose an energy-efficient, privacy-preserving solution based on an RNS-FHE scheme, specifically the HoRNS-CNN model. The practical implication of our study is that proposed RNS-FHE scheme allows for arbitrary homomorphic computations within deep CNN models to be performed directly on neuroimaging datasets with minimal computational overhead and manageable cipher expansion. In summary, our approach integrates an energy-efficient, privacy-preserving scheme to protect medical datasets in the cloud, addressing the global health crisis and the challenges of data sharing among institutions. This method not only maintains high accuracy in dyslexia neural-biomarker classification task but also ensures the privacy and security of the neuroimaging datasets used in all our experiments. Future research should develop a comprehensive strategy and algorithm to enable the proposed HoRNS-CNN model to consistently maintain an optimal balance between classification accuracy and energy efficiency, ensuring its practical deployment in real-world medical applications. By refining FHE parameters, incorporating lightweight CNN architectures, and implementing efficient computational techniques, the model can achieve high performance while minimizing energy consumption. This advancement will enhance the feasibility and sustainability of privacy-preserving neural biomarker classification.

All the results were generated experimentally in the encrypted domain. Our proposal enhances existing concepts such as CryptoNets and CryptoDL, in which DL algorithms are directly applied to cipher-images based on the principles of homomorphic privacy-preserving classification. In the future, we plan to implement more robust and complex models on recent GPU

processors, as well as investigate deep neural network privacy-preserving feature extraction timing performance in addition to classification performance. We also plan to take a comprehensive and holistic approach to dealing with hyperparameter optimization problems and augmentation in the encrypted domain to achieve state-of-the-art performance comparable to plain networks. One of the major limitations of this study is the scope of neuro-developmental disability covered. The study is primarily concerned with identifying dyslexia neural-biomarkers from cipher-images using the proposed HoRNS-CNN model. Other learning disabilities' neural-biomarkers, such as autism spectrum disorder (ASD) and attention deficit hyperactivity disorder (ADHD), were not considered. Second, due to the nature of the neuroimaging datasets used, this study was limited to a binary classification task only. Future research should consider the multi-class approach. Because dyslexia is a heterogeneous learning disorder with heterogeneous symptoms, the suggestion becomes appropriate [67]. Future studies could take advantage of the heterogeneity to develop a multi-class, robust homomorphic privacy-preserving model for its neural-biomarkers analysis. Finally, CTOPP scores were used to categorize subjects whose neuroimages were used in this study due to their effectiveness in measuring elements of phonological processing, which are hallmarks of dyslexia identification [53], but no single assessment may be sufficient in diagnosing dyslexia. Future research should compare the categorization abilities of other psycho-educational tests to the current study to see if a positive correlation exists.

Supplementary Information

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Supplementary Material 1.

Supplementary Material 2.

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Author contributions

Conceptualization: O.L.U.; Data Curation: O.L.U.; Formal Analysis: O.L.U., R.C.M.; Funding Acquisition: R.C.M.; Investigation: O.L.U.; Methodology: O.L.U., R.C.M., K.O., M.M., A.A.O., M.A.K.; Project Administration: R.C.M.; Resources: O.L.U., R.C.M., K.O., M.M., A.A.O., M.A.K.; Software: O.L.U., R.C.M.; Supervision: R.C.M.,

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Data availability

The raw datasets supporting all experiments and conclusions of this article are available on the OpenNeuro and C-BIRDS Databases and can be accessed via the links below: i. <https://openneuro.org/datasets/ds002236/versions/1.1.0> and ii. http://dx.doi.org/https://doi.org/10.15387/fcp_indi.corr.bnu1.

Declarations

Ethics approval and consent to participate

This article does not contain any studies with human participants or animals performed by any of the authors.

Competing interests

The authors declare no competing interests.

Author details

¹Research Centre for Cyber Security, Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia (UKM), 43600 Bangi, Selangor, Malaysia. ²Department of Computer Science, Tai Solarin University of Education, P.M.B. Ijagun, Ijebu-Ode 2118, Ogun, Nigeria. ³Research Centre for Artificial Intelligence Technology, Faculty of Information Science and Technology, Universiti Kebangsaan Malaysia (UKM), 43600 Bangi, Selangor, Malaysia. ⁴Center for Diagnostic, Therapeutic and Investigative Studies, Faculty of Health Sciences, Universiti Kebangsaan Malaysia, Jalan Raja Muda Abdul Aziz, Kuala Lumpur, Malaysia.

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References

- Alex S, Dhanaraj KJ, Deepthi PP (2022) Private decision tree-based disease detection with energy-efficiency at resource-constrained medical user in mobile healthcare network. *IEEE Access*. <https://doi.org/10.1109/ACCESS.2022.3149771>
- Tanuwidjaja HC, Choi R, Baek S, Kim K (2020) Privacy-preserving deep learning on machine learning as a service-a comprehensive survey. *IEEE Access* 8:167425–167447
- Boulemtafes A, Derhab A, Challal Y (2020) A review of privacy-preserving techniques for deep learning. *Neurocomputing* 384:21–45
- Usman OL, Muniyandi RC (2020) CryptoDL: predicting dyslexia biomarkers from encrypted neuroimaging dataset using energy-efficient residue number system and deep convolutional neural network. *Symmetry (Basel)* 12:1–24
- Tayyab M et al (2021) Cryptographic based secure model on dataset for deep learning algorithms. *Comput Mater Contin* 69:1183–1200
- Eichelberg M, Kleber K, Kämmerer M (2020) Cybersecurity in PACS and medical imaging: an overview. *J Digit Imaging* 33:1527–1542
- Hasan MK et al (2023) Review on cyber-physical and cyber-security system in smart grid: standards, protocols, constraints, and recommendations. *J Netw Comput Appl* 209:1–23
- Usman LO, Chandren R, Morufat M, Usman A (2023) Efficient neuroimaging data security and encryption using pixel - based homomorphic residue number system. *SN Comput Sci* 4:1–10
- Kwabena OA, Qin Z, Zhuang T, Qin Z (2019) MSCryptoNet: multi-scheme privacy-preserving deep learning in cloud computing. *IEEE Access* 7:29344–29354
- Usman OL, Muniyandi RC, Omar K, Mohamad M. Privacy-preserving classification method for neural-biomarkers using homomorphic residue number system CNN: HoRNS-CNN. In: 2022 International Conference on Business Analytics for Technology and Security, ICBATS 2022 (IEEE, 2022). <https://doi.org/10.1109/ICBATS54253.2022.9759007>.
- Gatta MT, Al-Latif STA (2018) Medical image security using modified chaos-based cryptography approach. *J Phys Conf Ser* 1003:1–6
- Al-Haj A, Abandah G, Hussein N (2015) Crypto-based algorithms for secured medical image transmission. *IET Inf Secur* 9:365–373
- Dowlin N et al (2016) Cryptonets: applying neural networks to encrypted data with high throughput and accuracy. *ICML* 1:342–351
- Al Badawi A et al (2021) The AlexNet moment for homomorphic encryption: HCNN, the first homomorphic CNN on encrypted data with GPUs. *IEEE Trans Emerg Top Comput* 9:1–13
- Chao J, et al. CaREnets: compact and resource-efficient CNN for homomorphic inference on encrypted medical images. *arXiv:1901.10074v1* 1–11, 2019.
- Muhammed KJ, Isiaka RM, Asaju-Gbolagade AW, Adewole KS, Gbolagade KA. Improved cloud-based N-primes model for symmetric-based fully homomorphic encryption using residue number system. In: Machine Learning and Data Mining for Emerging Trend in Cyber Dynamics (eds. Chiroma, H., Abdulhamid, S M Fournier-Viger, P. & Garcia, N. M.) 197–216 (Springer, Cham., 2021). https://doi.org/10.1007/978-3-030-66288-2_8.
- Gentry C. A fully homomorphic encryption scheme. Stanford University (Stanford University, 2009).
- Chabanne H, Wargny A de, Milgram J, Morel C, Prouff E. Privacy-preserving classification on deep neural network. In: IACR Cryptology ePrint Archive 35–48, 2017.
- Hesamifard E, Takabi H, Ghasemi M. CryptoDL: deep neural networks over encrypted data. *arXiv:1711.05189v1* 1–21, 2017.
- Vizitiu A, Năă CI, Puiu A, Suciu C, Itu LM. Applying deep neural networks over homomorphic encrypted medical data. *Comput Math Methods Med*. 2020, 2020.
- Chou, E. et al. Faster CryptoNets-leveraging sparsity for real-world encryption inference. *arXiv:1811.09953v1*, 2018.
- Chuman T, Sirichotedumrong W, Kiya H (2019) Encryption-then-compression systems using grayscale-based image encryption for JPEG images. *IEEE Trans Inf Forensics Secur* 14:1515–1525
- Itier V, Puteaux P, Puech W (2020) Recompression of jpeg crypto-compressed images without a key. *IEEE Trans Circuits Syst Video Technol* 30:646–660
- Maekawa T, Kawamura A, Kinoshita Y, Kiya H. Privacy-preserving SVM computing in the encrypted domain. In: 2018 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference, APSIPA ASC 2018 - Proceedings 897–902, 2018. <https://doi.org/10.23919/APSIPA.2018.8659529>.
- Tanaka M. Learnable image encryption. In: IEEE Conference on Consumer Electronics-Taiwan (ICCE-TW) 1–2 (2018).
- Li J, Kuang X, Lin S, Ma X, Tang Y (2020) Privacy preservation for machine learning training and classification based on homomorphic encryption schemes. *Inf Sci (Nij)* 526:166–179
- Ko DH, Choi SH, Shin JM, Liu P, Choi YH (2020) Structural image de-identification for privacy-preserving deep learning. *IEEE Access* 8:119848–119862
- Kurihara K, Imaizumi S, Shiota S, Kiya H (2017) An Encryption-then-compression system for lossless image compression standards. *IEICE Trans Inf Syst* E100D:52–56
- Martins P, Sousa L. A stochastic number representation for fully homomorphic cryptography. in: 2017 IEEE International Workshop on Signal Processing Systems (SiPS) 1–6 (IEEE, 2017). <https://doi.org/10.1109/SiPS.2017.8109973>.
- Sirichotedumrong W, Maekawa T, Kinoshita Y, Kiya H. Privacy-preserving deep neural networks with pixel-based image encryption considering data augmentation in the encrypted domain. In: IEEE International Conference on Image Processing 1–5, 2019.
- Sun X, Zhang P, Liu JK, Yu J, Xie W (2020) Private machine learning classification based on fully homomorphic encryption. *IEEE Trans Emerg Top Comput* 8:352–364
- Takabi H, Hesamifard E, Ghasemi M. Privacy Preserving Multi-party Machine Learning with Homomorphic Encryption. *Proc Work Priv Multi-Party Mach. Learn.* 1–5, 2016.
- Gomathisankaran M, Namuduri K, Tyagi A (2013) HORNS: a semi-perfectly secret homomorphic encryption system. *Am J Sci Eng* 2:17–23

34. Kong Y, Asif S, Khan MAU (2016) Modular multiplication using the core function in the residue number system. *Appl Algebr Eng Commun Comput* 27:1–16
35. Montgomery PL (1985) Modular multiplication without trial division. *Math Comput* 44:519–521
36. Moetesum M, Siddiqi I, Ehsan S, Vincent N (2020) Deformation modeling and classification using deep convolutional neural networks for computerized analysis of neuropsychological drawings. *Neural Comput Appl* 32:12909–12933
37. Rivest RL, Shamir A, Adleman L (1978) A method for obtaining digital signatures and public-key cryptosystems. *Commun ACM* 21:120–126
38. Chillotti I, Gama N, Georgieva M, Izabachène M. Faster fully homomorphic encryption: Bootstrapping in less than 0.1 seconds. in *Advances in Cryptology – ASIACRYPT 2016*. ASIACRYPT 2016. Lecture Notes in Computer Science (eds. Cheon, J. & Takagi, T.) vol. 10031 LNCS 3–33 (Springer, 2016).
39. Gentry C, Halevi S. Implementing Gentry's Fully-Homomorphic Encryption Scheme. 1–29, 2011.
40. Sharma I. Fully Homomorphic Encryption Scheme with Symmetric Keys. (University College of Engineering, Rajasthan Technical University, Kota, 2013).
41. Muhammed KJ, Gbolagade KA. Enhanced MORE Algorithm for Fully Homomorphic Encryption Based on Secret Information Moduli Set. BT - 2019 IEEE International Conference on Industrial Engineering and Engineering Management, IEEM 2019, Macao, Macao, December 15–18, 2019. 469–473 at <https://doi.org/10.1109/IEEM44572.2019.8978501>, 2019.
42. Youssef MI, Emam AE, Saafan SM, Elghany MABD (2013) Secured image encryption scheme using both residue number system and DNA sequence. *Online J Electron Electr Eng* 6:656–664
43. Alhassan S, Gbolagade KA (2013) Enhancement of the security of a digital image using the moduli set. *Int J Adv Res Comput Eng Technol* 2:2223–2229
44. Weyori BA, Amponsah PN, Yeboah PK (2012) Modeling a secured digital image encryption scheme using a three moduli set. *Glob J Comput Sci Technol Interdiscip* 12:1–9
45. Usman OL, Olusanya OO, Adedeji OB, Rufai KI (2018) Modelling a secure digital image cryptosystem using the traditional moduli set. *TASUED J Pure Appl Sci* 1:197–207
46. Younes D, Steffan P. A Comparative Study on Different Moduli Sets in Residue Number System. In: 2012 International Conference on Computer Systems and Industrial Informatics 1–6 (IEEE, 2012). <https://doi.org/10.1109/ICCSII.2012.6454344>.
47. Usman OL, Muniyandi RC, Omar K, Mohamad M (2021) Gaussian smoothing and modified histogram normalization methods to improve neural-biomarker interpretations for dyslexia classification mechanism. *PLoS ONE* 16:1–27
48. Lytle, M. N., Bitan, T. & Booth, J. R. A neuroimaging dataset on orthographic, phonological and semantic word processing in school-aged children. *Data Br.* **28**, 0–7 (2020).
49. Gorgolewski KJ et al (2016) The brain imaging data structure, a format for organizing and describing outputs of neuroimaging experiments. *Sci Data* 3:208–208
50. Haight, S. L. Test Review: Wagner, R.K., Torgesen, J.K., & Rashotte, C.A. (1999). *Comprehensive Test of Phonological Processing (CTOPP)*. Austin, TX:PRO-ED. *Assessment Eff. Interv.* **31**, 81–84 (2006).
51. Zhang Y, Brady M, Smith S (2001) Segmentation of brain MR images through a hidden Markov random field model and the expectation-maximization algorithm. *IEEE Trans Med Imaging* 20:45–57
52. Andersson JLR, Jenkinson M, Smith S. Non-linear registration, aka spatial normalization: FMRIB Technical Report TR07JA2. 2007.
53. van der Mark S et al (2011) The left occipitotemporal system in reading: Disruption of focal fMRI connectivity to left inferior frontal and inferior parietal language areas in children with dyslexia. *Neuroimage* 54:2426–2436
54. Usman OL, Muniyandi RC (2020) A framework for a secure brain image classification using deep learning and residue number system. *TEST Eng Manag* 83:6323–6330
55. Gbolagade KA, Cotofana SD. An O(n) Residue Number System to Mixed Radix Conversion Technique. In: IEEE Conference on Very Large Scale Integration 521–524 (2009). <https://doi.org/10.1109/ISCAS.2009.5117800>.
56. Gazizullina A. Fully homomorphic encryption scheme for secure computation. In: *ACM 2nd International Conference on Art, Science, and Engineering of Programming* 224–226 (ACM, 2018).
57. Pham V-T, Tran T-T, Wang P-C, Chen P-Y, Lo M-T (2021) EAR-UNet: a deep learning-based approach for segmentation of tympanic membranes from otoscopic images. *Artif Intell Med* 115:1–12
58. Obla S, Gong X, Aloufi A, Hu P, Takabi D (2020) Effective activation functions for homomorphic evaluation of deep neural networks. *IEEE Access* 8:153098–153112
59. Liu J, Liu B, Fu H. Optimizing Residue Number System on FPGA. in *Proceedings - 2016 IEEE International Conference on Internet of Things; IEEE Green Computing and Communications; IEEE Cyber, Physical, and Social Computing; IEEE Smart Data, iThings-GreenCom-CPSCo-Smart Data 2016* 621–624 (IEEE, 2016). <https://doi.org/10.1109/iThings-GreenCom-CPSCo-SmartData.2016.137>.
60. Lu B et al (2022) A practical Alzheimer's disease classifier via brain imaging-based deep learning on 85,721 samples. *J Big Data.* <https://doi.org/10.1186/s40537-022-00650-y>
61. Płoński P et al (2017) Multi-parameter machine learning approach to the neuroanatomical basis of developmental dyslexia. *Hum Brain Mapp* 38:900–908
62. Zahia S, Garcia-Zapirain B, Saralegui I, Fernandez-Ruanova B (2020) Dyslexia detection using 3D convolutional neural networks and functional magnetic resonance imaging. *Comput Methods Programs Biomed* 197:1–9
63. Tamboer P, Vorst HCM, Ghebreab S, Scholte HS (2016) Machine learning and dyslexia: classification of individual structural neuro-imaging scans of students with and without dyslexia. *Neuroimage Clin* 11:508–514
64. Aldehim G, Rashid M, Alluhaidan AS, Sakri S, Basheer S (2024) Deep learning for dyslexia detection: a comprehensive CNN approach with handwriting analysis and benchmark comparisons. *J Disabil Res* 3:1–8
65. Rouhani BD, Riazzi MS, Koushanfar F. DeepSecure: Scalable Provably-Secure Deep Learning. in *DAC '18: Proceedings of the 55th Annual Design Automation Conference* 1–6 (Association for Computing Machinery, 2018). <https://doi.org/10.1145/3195970.3196023>.
66. Sitaula C, Xiang Y, Aryal S, Lu X. Unsupervised deep features for privacy image classification. *arXiv:1909.10708v1* 1–12, 2019.
67. Usman OL, Muniyandi RC, Omar K, Mohamad M (2021) Advance Machine learning methods for dyslexia biomarker detection: a review of implementation details and challenges. *IEEE Access* 9:36879–36897

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